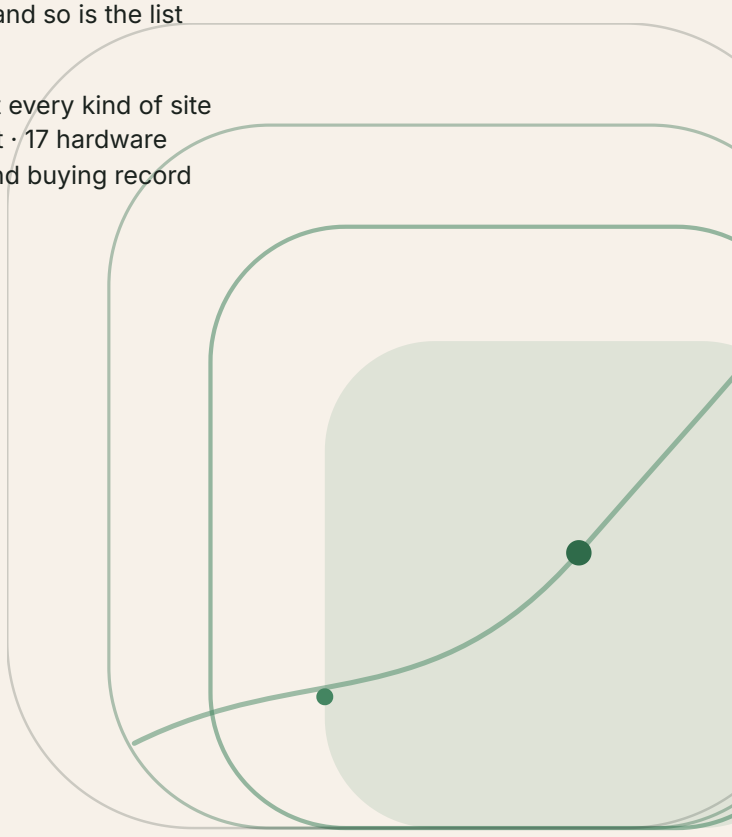
Greecon
SH.P.K

GUIDE 10 · EVERY PART THAT QUALIFIES

Qualified Hardware Catalogue

Every part that qualifies for a Greecon kit, one sheet each: how it qualifies, what it does at each of the four kinds of site, its settings, how it is fitted and by whom, how to accept it, and what goes wrong.

FOR	The founder and the project manager who buy hardware, and the electrician who fits it
PRINT	Once, to keep in the office with the quotes. Each sheet is also a PDF of its own (HW01 to HW17), and so is the list for each kind of site
COVERS	How a part qualifies · every part against every kind of site · the kit of each kind of site, part by part · 17 hardware sheets, each with its acceptance test and buying record



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Qualified hardware

This catalogue is every kind of hardware that qualifies for a Greecon kit, with one sheet for each. A sheet says how the part qualifies, what it does at each of the four kinds of site, how it is set and by whom it is fitted, how to accept it, and what goes wrong. Each sheet is also a PDF of its own, so that the one sheet a job needs can be printed and taken to the supplier, to the desk or to the site.

STOP

Greecon only reads, and only an electrician works on mains. No part in this catalogue switches, controls or protects anything. Everything inside a distribution board, every current clamp and voltage tap, and the pump's running contact are the electrician's work, done under the customer's lock-out. Greecon is not a safety system: the float switches, relief valves and trips already in the customer's equipment stay the safety layer.

How hardware qualifies

A part qualifies in one of three ways. The way decides what is bought, and what a person has to do in the platform.

A part qualifies as	Which means	In the platform	The sheets
Approved model	A named make and model. Greecon knows its registers, its serial settings and how to recognise it on a network. The model named is the model bought: a look-alike is not the same part.	The box finds it when a person presses Find equipment, and its readings are mapped already. Nobody types an address or a register.	HW05 Eastron SDM630 three-phase meter; HW06 Eastron SDM230 single-phase meter; HW08 Waveshare 4 to 20 mA module; HW09 Waveshare pump contact module
Qualified by specification	Any make that meets every line of the specification on its sheet. The specification is what matters, not the brand, so the part can be changed when a maker stops selling it.	The platform never talks to it by name. A transmitter is read through the 4 to 20 mA module and is asked for one thing, the range on its label. The others carry power, data or shelter.	HW01 Box; HW02 Converter; HW03 Data cable and labels; HW04 Cellular router; HW10 Level transmitter; HW11 Pressure transmitter; HW12 Flow meter; HW15 24 V power supply; HW16 UPS; HW17 Enclosure and fittings
Qualified by its register table	A Modbus device that is not an approved model. It qualifies only if its maker publishes its register table. Without the table it cannot be mapped honestly, so it is not bought, or not read.	The box cannot recognise it on a network. Its address is entered by hand, and each reading is mapped by hand from the maker's table.	HW07 Battery system; HW13 Soil moisture probe; HW14 Temperature and humidity sensor

A kit is made of approved models for what is read over the data cable, and of parts bought by specification for the rest. A soil probe, a weather sensor and a battery are the three that qualify by their maker's register table: if the maker will not give the table, the part is not bought, or not read.

Every part at every kind of site

The four kinds of site are **H**, a hotel or guesthouse (water, solar and grid); **S**, a solar plant with a grid connection; **F**, a farm or greenhouse with irrigation; and **W**, a water tank and pump station. A number is how many the standard kit has. "0 or 1" is a part that is fitted only where the site calls for it. A dash is a part that kind of site does not use.

Sheet	Hardware	Qualifies as	H	S	F	W	Planning estimate (EUR)
HW01	**Box**	By specification	1	1	1	1	180 to 300
HW02	**Converter**	By specification	1	1	1	1	30 to 60 (basic) or 250 and up (industrial)
HW03	**Data cable and labels**	By specification	1 lot	1 lot	1 lot	1 lot	40 to 80
HW04	**Cellular router**	By specification	0 or 1	0 or 1	0 or 1	0 or 1	150 to 200
HW05	**Eastron SDM630 three-phase meter**	Approved model	2	2, or 3	–	–	90 to 180 each
HW06	**Eastron SDM230 single-phase meter**	Approved model	–	–	–	0 or 1	35 to 60
HW07	**Battery system**	By register table	–	0 or 1	–	–	Nothing to buy: it is the customer's own
HW08	**Waveshare 4 to 20 mA module**	Approved model	1	–	1	1	30 to 60
HW09	**Waveshare pump contact module**	Approved model	1	–	0 or 1	1	30 to 60
HW10	**Level transmitter**	By specification	1	–	1	1	120 to 400
HW11	**Pressure transmitter**	By specification	1	–	1	1	40 to 150
HW12	**Flow meter**	By specification	0 or 1	–	0 or 1	0 or 1	150 to 500
HW13	**Soil moisture probe**	By register table	–	–	1 for each zone	–	40 to 120 each
HW14	**Temperature and humidity sensor**	By register table	–	–	1	–	20 to 80
HW15	**24 V power supply**	By specification	1	1	1	1	25 to 50
HW16	**UPS**	By specification	1	1	–	1	60 to 150
HW17	**Enclosure and fittings**	By specification	1	1	1	1	80 to 150

The planning figure for a whole kit, without VAT and without labour:

Kind of site	Planning figure
H Hotel or guesthouse	About EUR 800 to 1,800; typically 1,100 to 1,300. Add the electrician's time (half a day to a day) and yours.
S Solar plant with grid connection	About EUR 600 to 1,200.
F Farm or greenhouse with irrigation	About EUR 700 to 1,500.
W Water tank and pump station	About EUR 650 to 1,300.

The four lists after this chapter read the same catalogue by kind of site: every part of that kit, what it does there, and its sheet.

The numbers that never change

A new site looks like the last one, because the same settings are used everywhere. Every number is set at the office, before the kit leaves, and only checked on site.

Number	Device	What it is	Sheet
1	Grid meter	Eastron SDM630, where the supply enters. An Eastron SDM230 on a single-phase supply	HW05 and HW06
2	Solar meter	Eastron SDM630, on the inverter's output. An Eastron SDM230 on a single-phase plant	HW05 and HW06
3	4 to 20 mA module	Waveshare Modbus RTU Analog Input 8CH. The platform calls it the Water I/O module	HW08
4	Load meter, or pump meter	Eastron SDM630 on the site's own consumption, or Eastron SDM230 on a single-phase pump	HW05 and HW06
5	Pump contact module	Waveshare Modbus RTU IO 8CH	HW09
6	Soil probe	Any probe that qualifies by its register table. A second probe takes the next free number, 8	HW13
7	Weather sensor	Any temperature and humidity sensor that qualifies by its register table	HW14

Setting	Value
Speed of everything on the data cable	9600 baud, 8 data bits, no parity, 1 stop bit
The converter	Modbus TCP to RTU gateway mode, port 502, a fixed address on the site's network
The transmitters	Level on input 1 of the 4 to 20 mA module, pressure on input 2, the flow meter on input 3
The pump's running contact	Input 1 of the pump contact module
What the box looks for	Port 502 on the site's own network, and unit numbers 1 to 10 behind whatever answers

Before you buy

1. **One of each first.** Buy one whole kit and rehearse it on a desk before buying five. Change the specification, not the kit, when something fails.
2. **Three quotes.** For an approved model, the model named is the model bought: a look-alike is not the same part. For everything else it is the specification, not the brand. Keep the quotes. Every sheet ends with a buying record.
3. **Keep a hardware register:** for each part the make and model, the supplier, the date, the serial number or the batch, and the invoice. A spreadsheet is enough.
4. **Prepare at the office.** The box has its software installed and tested, every device has its number and its speed, and the converter is set, before the kit leaves.

STOP

Do not buy: a meter or a module that is not an approved model; a probe, a sensor or a battery whose register table you have not seen; a transmitter that is not rated for the fluid; a box that starts from an SD card; or anything with a wireless link into a board that you cannot inspect.

How to read a sheet

Every sheet has the same parts, in the same order.

Part of the sheet	What it tells you
At a glance	What the part does, how it qualifies, its number on the cable, who fits it, the planning estimate and which kits use it
How it qualifies	The list to tick before buying, and what not to buy
At each kind of site	Whether a hotel, a solar plant, a farm and a water station have it, how many, and what it does there
What Greecon reads from it	The readings, their units and the standard alerts
Settings, and where they are made	What is set, to what, and whether at the office or on site
Fitting	Who does what. Mains work is always the electrician's

Part of the sheet	What it tells you
In the platform	What a person presses and types, by the names on the screen
Acceptance test	What is checked at the office, and what is checked on the day
What goes wrong	What you see, the likely cause, and what to do
Not yet proven	What has not yet been shown on real hardware
Buying record	Three quotes, and what was bought

Where a sheet is silent, the maker's manual decides. The drawings show how parts connect, not terminal numbers: those differ between makes and versions.

What is not yet proven

This catalogue is written before the first real kit. It says so on every sheet, and here once for all of them.

- **This software has not yet read a real meter, module, probe, sensor or battery.** The registers of the Eastron meters and the Waveshare modules are from the makers' published documents, and everything is tested against pretend equipment that answers as those documents say.
- **The first real kit is rehearsed on a desk.** Anything that differs from a sheet is written down and reported to Greecon, and the sheet is corrected.
- **The prices are planning figures** from shop listings seen in October 2026, not quotes.
- **The electrical parts of every sheet are to be read by a licensed electrician before they are used.** A sheet says who does the work. It does not replace the electrician's own judgement, the maker's manual or the law.

KIT H

Hardware for a hotel or guesthouse

A hotel or a guesthouse with a water tank and its pump, solar on the roof and a grid connection: the usual first site. Greecon measures what is bought from the grid and what is sent to it, what the solar plant produces, how full the tank is, the pressure in the pipe and, where the parts are fitted, the pump's flow and whether the pump runs. This list is every part of the kit for that, with the sheet that says how each one qualifies.

What is measured, and where

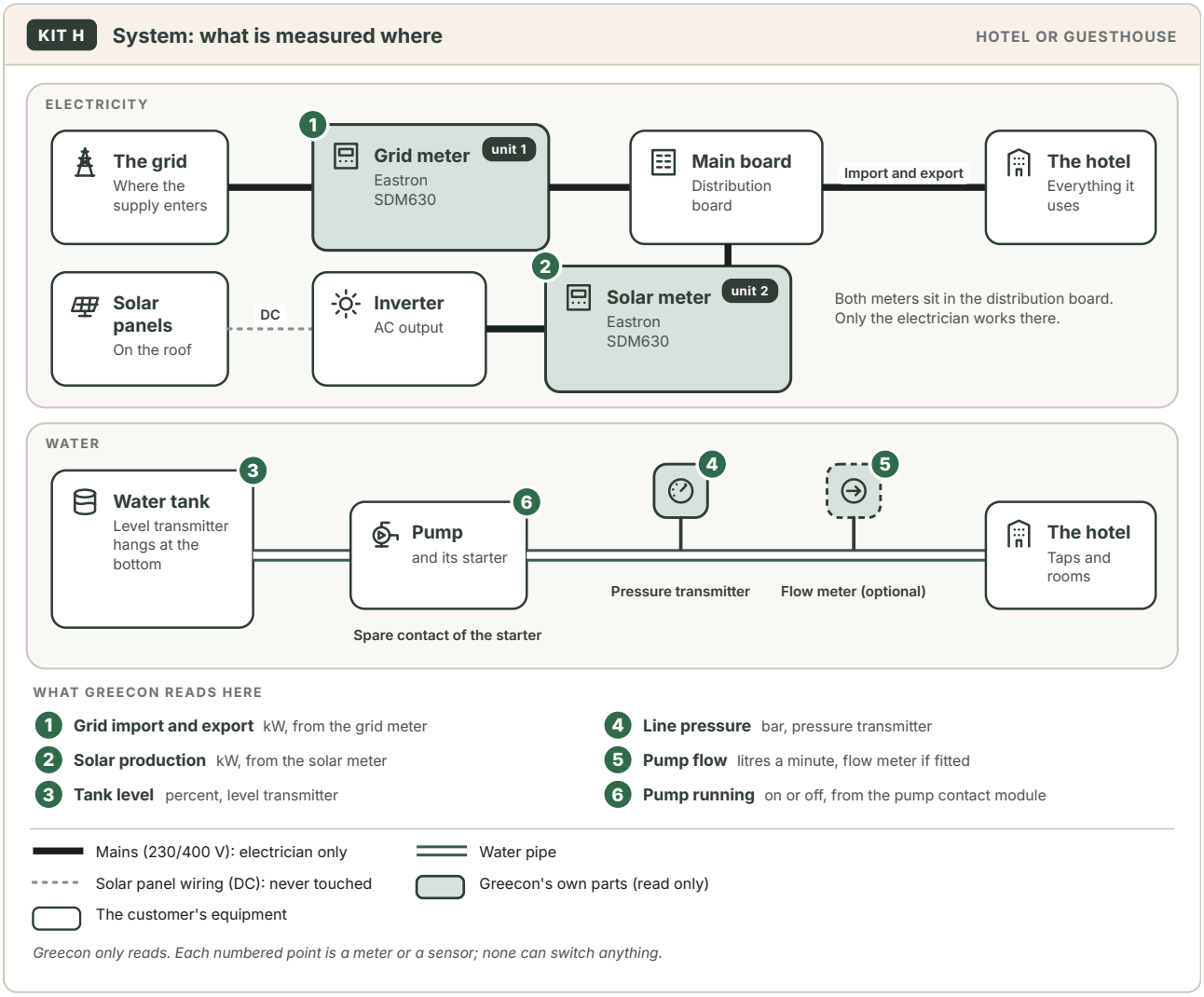


Figure H-1. Kit H, hotel or guesthouse. System: what is measured where. The numbers mark what Greecon reads; they are listed under the drawing.

The kit, part by part

Sheet	Hardware	In the kit	How many	What it does at this kit of site
HW01	**Box** By specification · EUR 180 to 300	IN THE KIT	1	In the plant room. It reads the grid meter, the solar meter, the water module and the pump contact module: four devices, seven readings.
HW02	**Converter** By specification · EUR 30 to 60 (basic) or 250 and up (industrial)	IN THE KIT	1	In the enclosure. Behind it, on one cable: the grid meter (number 1), the solar meter (2), the water module (3) and the pump contact module (5).
HW03	**Data cable and labels** By specification · EUR 40 to 80	IN THE KIT	1 lot	One run from the converter through the grid meter and the solar meter (in the board, where the electrician connects it) and the water module to the pump contact module.
HW04	**Cellular router** By specification · EUR 150 to 200	OPTIONAL	0 or 1	Only where the building has no internet the box may use. Most hotels have a router: then the box and the converter plug into that.
HW05	**Eastron SDM630 three-phase meter** Approved model · EUR 90 to 180 each	IN THE KIT	2	Two. The grid meter (number 1), where the supply enters, gives Grid import and Grid export. The solar meter (number 2), on the inverter's AC output, gives Solar production.
HW08	**Waveshare 4 to 20 mA module** Approved model · EUR 30 to 60	IN THE KIT	1	One. Input 1 takes the tank's level transmitter, input 2 the pipe's pressure transmitter, input 3 the flow meter if there is one. It gives Tank level, Line pressure and Pump flow.
HW09	**Waveshare pump contact module** Approved model · EUR 30 to 60	IN THE KIT	1	One: it reads a spare running contact of the water pump's starter and gives Pump running. Leave it out if the starter has no spare contact, and say so at handover.
HW10	**Level transmitter** By specification · EUR 120 to 400	IN THE KIT	1	One, at the bottom of the hotel's water tank. Only a transmitter with wetted parts approved for drinking water goes into a drinking-water tank. It gives Tank level.
HW11	**Pressure transmitter** By specification · EUR 40 to 150	IN THE KIT	1	One, on the pipe the pump feeds. It gives Line pressure.
HW12	**Flow meter** By specification · EUR 150 to 500	OPTIONAL	0 or 1	With the pump contact module it makes the no-flow alert possible: the pump is running but no water is flowing.

Sheet	Hardware	In the kit	How many	What it does at this kind of site
HW15	**24 V power supply** By specification · EUR 25 to 50	IN THE KIT	1	One, in the enclosure. It feeds the two modules, the transmitters' loops and the converter, and the box if the box takes 9 to 36 V.
HW16	**UPS** By specification · EUR 60 to 150	IN THE KIT	1	One. It keeps the box and the converter running through a short power cut. A 600 VA UPS keeps the 24 V supply up too, so the tank and the pressure are still read.
HW17	**Enclosure and fittings** By specification · EUR 80 to 150	IN THE KIT	1	One: the fuses and the 24 V supply on the mains side; the converter, the 4 to 20 mA module and the pump contact module on the low-voltage side; the terminal blocks and, if there is room, the box.

About EUR 800 to 1,800; typically 1,100 to 1,300. Add the electrician's time (half a day to a day) and yours. The figures are planning estimates, not quotes.

Sheet	Hardware	Why it is not in this kit
HW06	**Eastron SDM230 single-phase meter**	Not in the standard kit, which is three-phase. On a single-phase supply it takes the SDM630's place at the grid connection (number 1) and on the inverter's output (number 2). Find equipment says which model it found.
HW07	**Battery system**	The hotel template has no battery.

Never at this kind of site: HW13 Soil moisture probe; HW14 Temperature and humidity sensor.

The numbers on the data cable

Number	Device	What it is	Sheet
1	Grid meter	Eastron SDM630, where the supply enters. An Eastron SDM230 on a single-phase supply	HW05 and HW06
2	Solar meter	Eastron SDM630, on the inverter's output. An Eastron SDM230 on a single-phase plant	HW05 and HW06
3	4 to 20 mA module	Waveshare Modbus RTU Analog Input 8CH. The platform calls it the Water I/O module	HW08
5	Pump contact module	Waveshare Modbus RTU IO 8CH	HW09

Everything on the cable is set to 9600 baud, 8 data bits, no parity, 1 stop bit, and the converter to port 502, in Modbus TCP to RTU gateway mode. Every number is set at the office, before the kit leaves.

What the survey decides

Ask, or look	It decides
Is the supply three-phase, and how many amps a phase?	The meters. Up to 100 A a phase: the SDM630 MCT, with clamps. Above that: the SDM630 with 5 A current transformers. A single-phase supply: the SDM230 (sheets HW05 and HW06)
How tall is the tank inside, and does it hold drinking water?	The level transmitter’s range, and what it may be made of (sheet HW10)
What pressure does the pipe see?	The pressure transmitter’s range (sheet HW11)
Has the pump’s starter a spare contact that follows the pump?	Whether the pump contact module is fitted (sheet HW09)
Does the owner want to be told when the pump runs and no water flows?	The flow meter, which needs the pump’s contact as well (sheet HW12)
Is there internet the box may use, where the enclosure will be?	The cellular router (sheet HW04)
Where does the enclosure go, and is the place dry?	IP54 or IP65 (sheet HW17)

The whole job at this kind of site, from the survey to the handover, is in the site pack for a hotel or guesthouse (PDF 04).

KIT S

Hardware for a solar plant with grid connection

A solar plant with a grid connection, with or without a battery. Greecon measures what is bought from the grid and what is sent to it, and what the plant produces. Where the owner wants it, a third meter measures the site's own consumption; without it Greecon works the consumption out. A battery is the customer's own equipment, and is read only if its maker publishes its register table. This list is every part of the kit for that, with the sheet that says how each one qualifies.

What is measured, and where

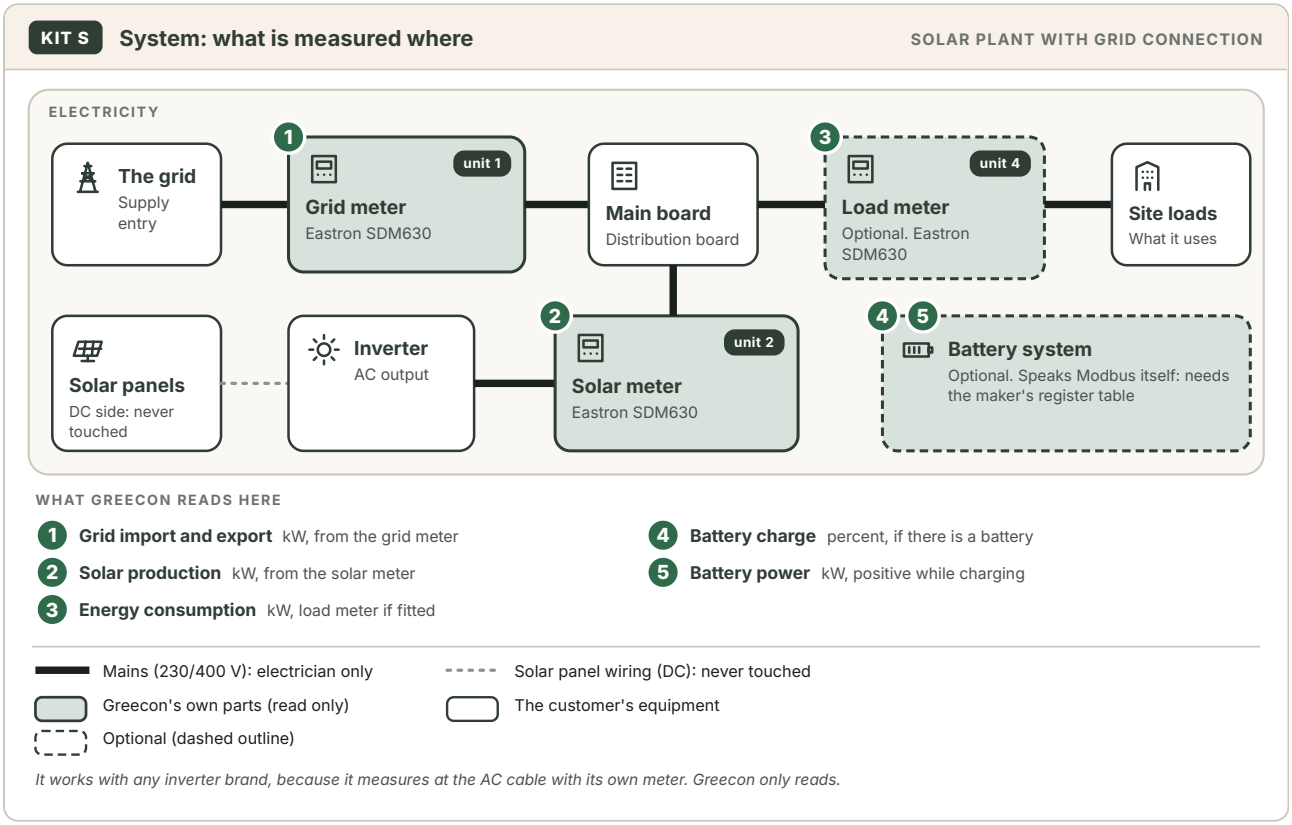


Figure S-1. Kit S, solar plant with grid connection. System: what is measured where. The dashed boxes are optional.

The kit, part by part

Sheet	Hardware	In the kit	How many	What it does at this kind of site
HW01	**Box** By specification · EUR 180 to 300	IN THE KIT	1	Near the main board or the inverter. It reads the grid meter and the solar meter and, where they are fitted, the load meter and the battery.
HW02	**Converter** By specification · EUR 30 to 60 (basic) or 250 and up (industrial)	IN THE KIT	1	In the enclosure. Behind it: the grid meter (1), the solar meter (2) and, if fitted, the load meter (4). A battery that speaks Modbus on its own network address does not need the converter; one on RS-485 joins the cable with a number of its own.
HW03	**Data cable and labels** By specification · EUR 40 to 80	IN THE KIT	1 lot	One run from the converter through the grid meter, the solar meter and, if fitted, the load meter. Every end at a meter is the electrician's.
HW04	**Cellular router** By specification · EUR 150 to 200	OPTIONAL	0 or 1	Only without usable internet. The hardware guide's own list for this kit does not show one; the network rules are the same as for every kit.
HW05	**Eastron SDM630 three-phase meter** Approved model · EUR 90 to 180 each	IN THE KIT	2, or 3	Two, as at a hotel: the grid meter (1) and the solar meter (2). A third (number 4) on the feed to the loads if the owner wants the site's own consumption measured and not worked out.
HW07	**Battery system** By register table · Nothing to buy: it is the customer's own	OPTIONAL	0 or 1	The template's Battery system (optional). Two readings are mapped by hand: Battery charge and Battery power. If the plant has no battery, remove the device.
HW15	**24 V power supply** By specification · EUR 25 to 50	IN THE KIT	1	One. With no module and no loop, it feeds the converter, and the box if the box takes 9 to 36 V.
HW16	**UPS** By specification · EUR 60 to 150	IN THE KIT	1	One. It keeps the box and the converter running. The meters are powered from what they measure: in a grid cut they go dark, and the box says that they do not answer.
HW17	**Enclosure and fittings** By specification · EUR 80 to 150	IN THE KIT	1	One, the smallest: the fuses and the supply, the converter, the terminal blocks and, if there is room, the box. The meters are in the distribution board, not here.

About EUR 600 to 1,200. The figures are planning estimates, not quotes. For a battery nothing is bought but the maker's register table.

Sheet	Hardware	Why it is not in this kit
HW06	**Eastron SDM230 single-phase meter**	As at a hotel: only on a single-phase plant, in the SDM630's place, with the same numbers (grid 1, solar 2, load 4).
HW08	**Waveshare 4 to 20 mA module**	A solar plant has no water side.
HW09	**Waveshare pump contact module**	A solar plant has no pump to watch.
HW10	**Level transmitter**	A solar plant has no water side.
HW11	**Pressure transmitter**	A solar plant has no water side.
HW12	**Flow meter**	A solar plant has no water side.

Never at this kind of site: HW13 Soil moisture probe; HW14 Temperature and humidity sensor.

The numbers on the data cable

Number	Device	What it is	Sheet
1	Grid meter	Eastron SDM630, where the supply enters. An Eastron SDM230 on a single-phase supply	HW05 and HW06
2	Solar meter	Eastron SDM630, on the inverter's output. An Eastron SDM230 on a single-phase plant	HW05 and HW06
4	Load meter, or pump meter	Eastron SDM630 on the site's own consumption, or Eastron SDM230 on a single-phase pump	HW05 and HW06

A battery that joins the data cable takes a number of its own, entered by hand (for example 8). One that is on the site's network at an address of its own has no number on the cable. Everything on the cable is set to 9600 baud, 8 data bits, no parity, 1 stop bit, and the converter to port 502, in Modbus TCP to RTU gateway mode.

What the survey decides

Ask, or look	It decides
Is the supply three-phase, and how many amps a phase?	The meters. Up to 100 A a phase: the SDM630 MCT, with clamps. Above that: the SDM630 with 5 A current transformers. A single-phase plant: the SDM230 (sheets HW05 and HW06)
Where are the supply entry and the inverter’s AC output?	Where the grid meter and the solar meter go, and how far the data cable runs (sheet HW03)
Does the owner want the site’s own consumption measured, and not worked out?	A third meter, number 4 (sheet HW05)
Is there a battery, and will its maker give the Modbus register table?	Whether the battery is read at all (sheet HW07)
May the plant send energy to the grid?	Nothing to buy. It is entered with the site’s prices, and it explains a solar reading that is low at midday
Is there internet the box may use, where the enclosure will be?	The cellular router (sheet HW04)

The whole job at this kind of site, from the survey to the handover, is in the site pack for a solar plant (PDF 05).

KIT F

Hardware for a farm or greenhouse with irrigation

A farm or a greenhouse with irrigation: the soil and the air at the field, with the water supply behind it. Greecon measures the moisture of the soil in each irrigation zone, the air temperature and humidity, how full the tank is, the pressure in the irrigation line and, if there is a pump to watch, the pump's flow and whether it runs. This list is every part of the kit for that, with the sheet that says how each one qualifies.

What is measured, and where

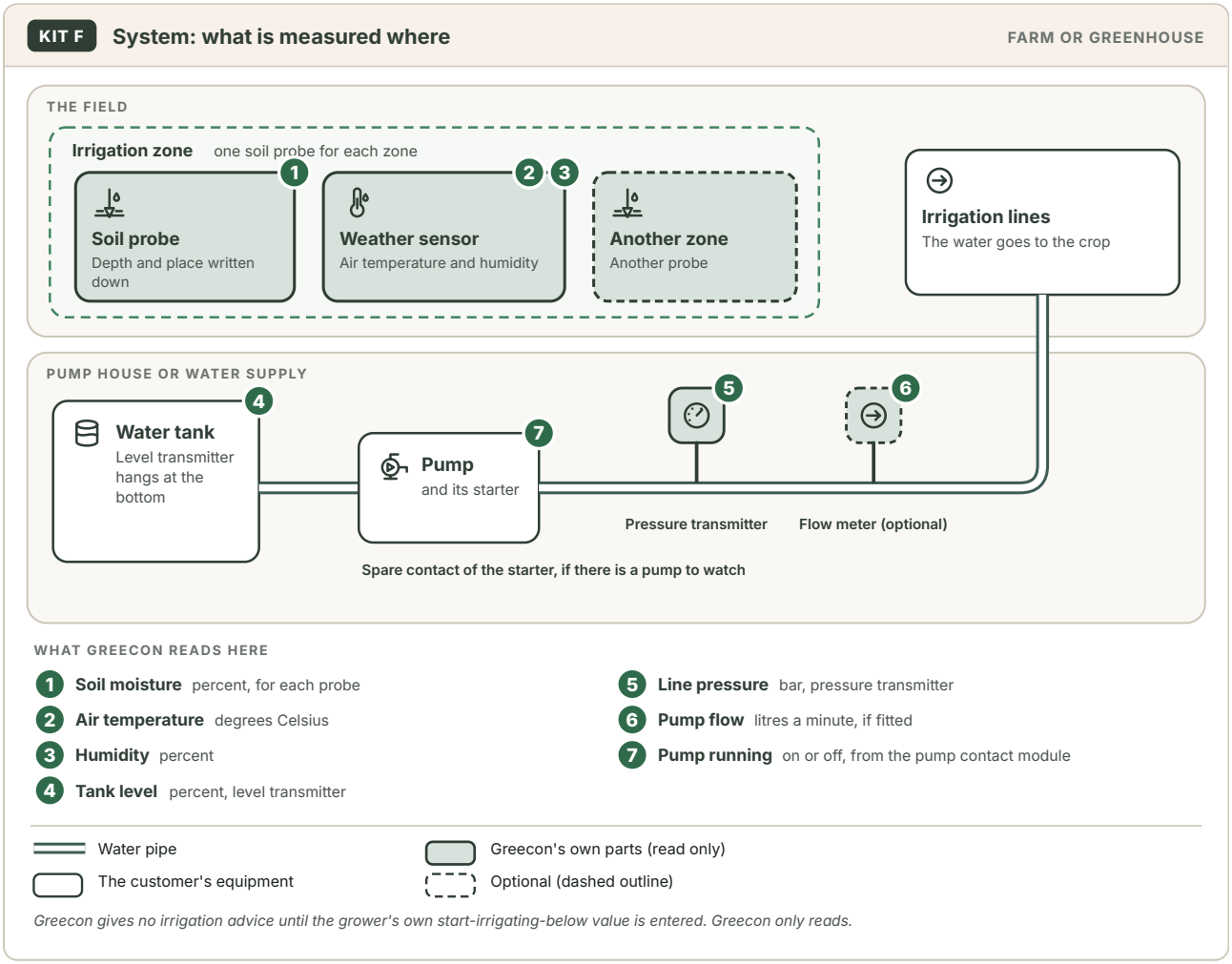


Figure F-1. Kit F, farm or greenhouse with irrigation. System: what is measured where. The dashed parts are optional.

The kit, part by part

Sheet	Hardware	In the kit	How many	What it does at this kind of site
HW01	**Box** By specification · EUR 180 to 300	IN THE KIT	1	In the pump house. It reads the soil probes, the weather sensor, the water module and, if there is a pump to watch, the pump contact module. A 4G router is the likelier way to the internet.
HW02	**Converter** By specification · EUR 30 to 60 (basic) or 250 and up (industrial)	IN THE KIT	1	In the enclosure in the pump house. Behind it: the water module (3), the pump contact module (5) if there is a pump to watch, the soil probe (6; a second probe takes 8) and the weather sensor (7).
HW03	**Data cable and labels** By specification · EUR 40 to 80	IN THE KIT	1 lot	The longest run of the four: from the converter in the pump house through the soil probe and the weather sensor in the field and the water module to the pump contact module. Plan it under 100 metres.
HW04	**Cellular router** By specification · EUR 150 to 200	IN THE KIT	0 or 1	The likelier case: a field or a pump house seldom has a cable to the internet. The box and the converter plug into the router's sockets.
HW08	**Waveshare 4 to 20 mA module** Approved model · EUR 30 to 60	IN THE KIT	1	One, for the tank and the pipe behind the irrigation. The flow meter (input 3) only if there is a pump to watch.
HW09	**Waveshare pump contact module** Approved model · EUR 30 to 60	OPTIONAL	0 or 1	Only if there is a pump to watch and its starter has a spare running contact.
HW10	**Level transmitter** By specification · EUR 120 to 400	IN THE KIT	1	One, in the irrigation tank or reservoir. It gives Tank level, with the standard low-level alerts.
HW11	**Pressure transmitter** By specification · EUR 40 to 150	IN THE KIT	1	One, on the irrigation line.
HW12	**Flow meter** By specification · EUR 150 to 500	OPTIONAL	0 or 1	Only if there is a pump to watch.
HW13	**Soil moisture probe** By register table · EUR 40 to 120 each	IN THE KIT	1 for each zone	One for each irrigation zone, at the depth and in the place agreed with the grower. It gives Soil moisture. Greecon gives no irrigation advice until the grower's own Start irrigating below value is entered.

Sheet	Hardware	In the kit	How many	What it does at this kind of site
HW14	**Temperature and humidity sensor** By register table · EUR 20 to 80	IN THE KIT	1	One, at the field or in the greenhouse. It gives Air temperature and Humidity. The frost alert says nothing until the grower's own Frost alert at or below value is entered.
HW15	**24 V power supply** By specification · EUR 25 to 50	IN THE KIT	1	One, as at a hotel: the modules, the loops and the converter.
HW17	**Enclosure and fittings** By specification · EUR 80 to 150	IN THE KIT	1	One, in the pump house: as at a hotel, with the 4G router on the rail. The probes and the weather sensor are in the field, on the data cable.

About EUR 700 to 1,500. The figures are planning estimates, not quotes. One soil probe is bought for each irrigation zone.

Sheet	Hardware	Why it is not in this kit
HW05	**Eastron SDM630 three-phase meter**	A farm kit has no energy meter. One can be added to any site from Add equipment Greecon already knows, and then this sheet applies.
HW06	**Eastron SDM230 single-phase meter**	A farm kit has no energy meter.
HW16	**UPS**	Not in the standard list for this kit. Decide at the survey whether this site gets one, and write the decision down.

Never at this kind of site: HW07 Battery system.

The numbers on the data cable

Number	Device	What it is	Sheet
3	4 to 20 mA module	Waveshare Modbus RTU Analog Input 8CH. The platform calls it the Water I/O module	HW08
5	Pump contact module	Waveshare Modbus RTU IO 8CH	HW09
6	Soil probe	Any probe that qualifies by its register table. A second probe takes the next free number, 8	HW13
7	Weather sensor	Any temperature and humidity sensor that qualifies by its register table	HW14

A second soil probe, for a second zone, takes the next free number: 8, then 9. The numbers of the probes and of the weather sensor are set **before the probe is buried and the sensor is mounted.** Everything on the cable is set to 9600 baud, 8 data bits, no parity, 1 stop bit, and the converter to port 502, in Modbus TCP to RTU gateway mode.

What the survey decides

Ask, or look	It decides
How many irrigation zones are there?	How many soil probes (sheet HW13)
Which crop, which soil, how deep are the roots, and which spot is typical of the zone?	Where each probe goes and how deep. The irrigation limit is the grower's own, and is asked after the first week
At what air temperature does frost damage the crop, and in which weeks?	The frost limit, and who is woken (sheet HW14)
How tall is the tank inside, and what pressure does the line see?	The ranges of the level and pressure transmitters (sheets HW10 and HW11)
Is there a pump to watch, and has its starter a spare contact that follows it?	The pump contact module and the flow meter (sheets HW09 and HW12)
Is there a phone signal at the pump house, or a network cable to it?	The cellular router, which is the likelier case in a field (sheet HW04)
How far is it from the pump house to the farthest probe?	The data cable: plan the whole run under 100 metres (sheet HW03)
Does this site get a UPS?	The standard list for this kit has none. Decide, and write the decision down (sheet HW16)

The whole job at this kind of site, from the survey to the handover, is in the site pack for a farm or greenhouse (PDF 06).

KIT W

Hardware for a water tank and pump station

A water tank, a pump and the pipe it feeds. Greecon measures how full the tank is, the pressure in the pipe, whether the pump runs and, where the parts are fitted, the pump's flow and the electricity the pump uses. With the flow meter and the pump's running contact together, Greecon can say that the pump is running but no water is flowing. This list is every part of the kit for that, with the sheet that says how each one qualifies.

What is measured, and where

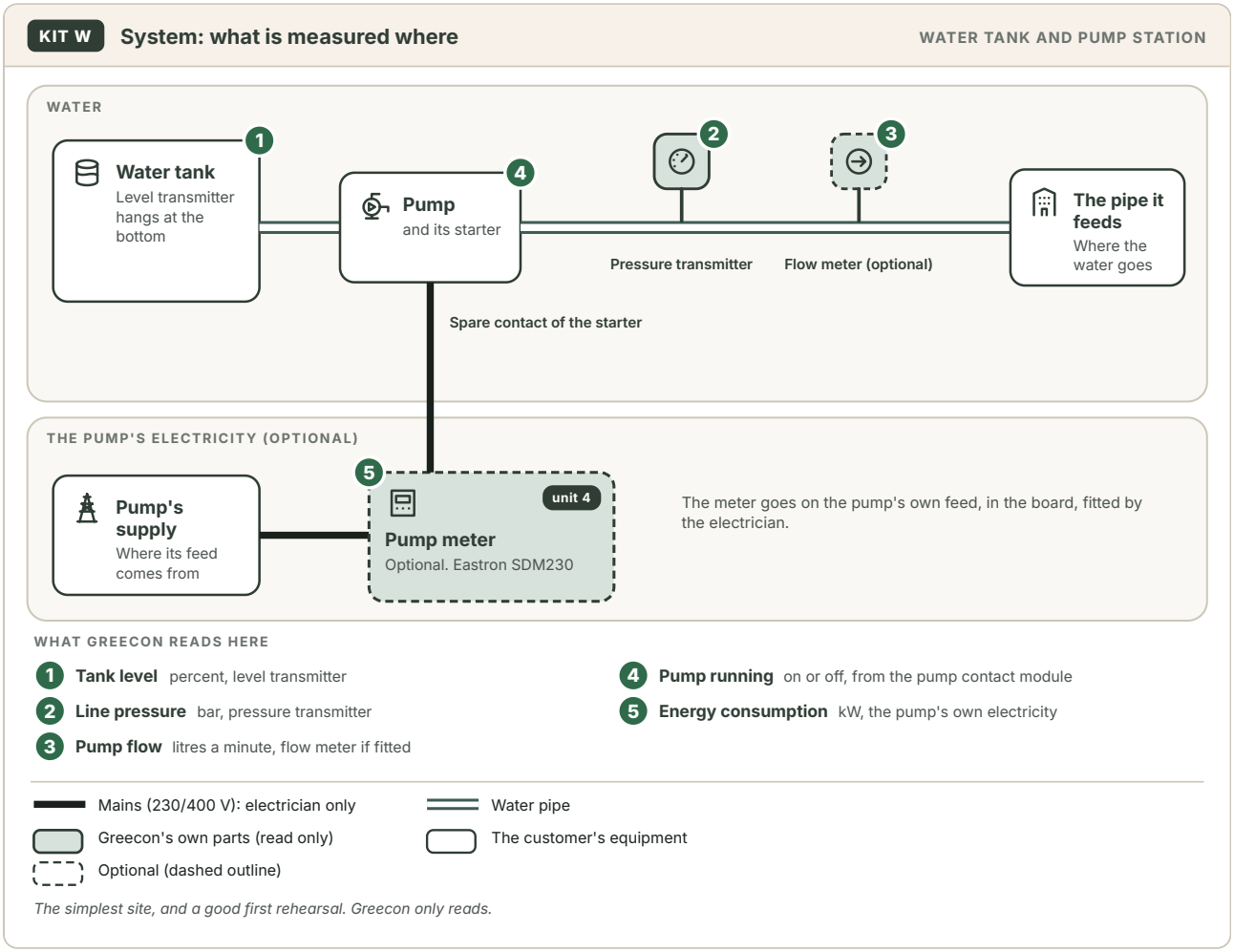


Figure W-1. Kit W, water tank and pump station. System: what is measured where. The dashed parts are optional.

The kit, part by part

Sheet	Hardware	In the kit	How many	What it does at this kind of site
HW01	**Box** By specification · EUR 180 to 300	IN THE KIT	1	In the pump house or the plant room. It reads the water module, the pump contact module and, if one is fitted, the pump meter.
HW02	**Converter** By specification · EUR 30 to 60 (basic) or 250 and up (industrial)	IN THE KIT	1	In the enclosure. Behind it: the water module (3), the pump contact module (5) and, if fitted, the pump meter (4).
HW03	**Data cable and labels** By specification · EUR 40 to 80	IN THE KIT	1 lot	A short run inside the pump house: from the converter through the pump meter (if fitted) and the water module to the pump contact module.
HW04	**Cellular router** By specification · EUR 150 to 200	OPTIONAL	0 or 1	Only without usable internet: a pump station that stands away from the buildings.
HW06	**Eastron SDM230 single-phase meter** Approved model · EUR 35 to 60	OPTIONAL	0 or 1	On the pump’s own feed, as the pump meter (number 4). It gives Energy consumption: the pump’s electricity. For a single-phase pump; a three-phase pump takes the SDM630 (sheet HW05).
HW08	**Waveshare 4 to 20 mA module** Approved model · EUR 30 to 60	IN THE KIT	1	One, the centre of the kit: level on input 1, pressure on input 2, the flow meter on input 3 if it is fitted.
HW09	**Waveshare pump contact module** Approved model · EUR 30 to 60	IN THE KIT	1	One, for the station’s pump. Leave it out if the starter has no spare running contact.
HW10	**Level transmitter** By specification · EUR 120 to 400	IN THE KIT	1	One, in the station’s tank. If the tank holds drinking water, the drinking-water rule applies.
HW11	**Pressure transmitter** By specification · EUR 40 to 150	IN THE KIT	1	One, on the pipe the pump feeds. The pipe’s pressure range, asked at the survey, decides the transmitter’s range.
HW12	**Flow meter** By specification · EUR 150 to 500	OPTIONAL	0 or 1	The part behind a pump station’s most useful alert: pump running, no flow. Without it the handover says that the alert cannot be raised.

Sheet	Hardware	In the kit	How many	What it does at this kind of site
HW15	**24 V power supply** By specification · EUR 25 to 50	IN THE KIT	1	One, as at a hotel: the modules, the loops and the converter.
HW16	**UPS** By specification · EUR 60 to 150	IN THE KIT	1	One. A 600 VA UPS is the better choice here: it keeps the 24 V supply up too, so the tank and the pressure are still read while the power is down.
HW17	**Enclosure and fittings** By specification · EUR 80 to 150	IN THE KIT	1	One: as at a hotel. The optional pump meter sits in the board that feeds the pump, not in this enclosure.

About EUR 650 to 1,300. The figures are planning estimates, not quotes.

Sheet	Hardware	Why it is not in this kit
HW05	**Eastron SDM630 three-phase meter**	The kit measures the pump's electricity with the single-phase SDM230 (sheet HW06). A three-phase pump needs this meter in its place: remove the template's pump meter and add Eastron SDM630 on the site's own consumption (number 4) from Add equipment Greecon already knows.

Never at this kind of site: HW07 Battery system; HW13 Soil moisture probe; HW14 Temperature and humidity sensor.

The numbers on the data cable

Number	Device	What it is	Sheet
3	4 to 20 mA module	Waveshare Modbus RTU Analog Input 8CH. The platform calls it the Water I/O module	HW08
4	Load meter, or pump meter	Eastron SDM630 on the site's own consumption, or Eastron SDM230 on a single-phase pump	HW05 and HW06
5	Pump contact module	Waveshare Modbus RTU IO 8CH	HW09

Everything on the cable is set to 9600 baud, 8 data bits, no parity, 1 stop bit, and the converter to port 502, in Modbus TCP to RTU gateway mode. Every number is set at the office, before the kit leaves.

What the survey decides

Ask, or look	It decides
How tall is the tank inside, and does it hold drinking water?	The level transmitter’s range, and what it may be made of (sheet HW10)
What pressure does the pipe see?	The pressure transmitter’s range (sheet HW11)
Has the pump’s starter a spare contact that follows the pump?	Whether the pump contact module is fitted (sheet HW09)
Does the owner want to be told when the pump runs and no water flows?	The flow meter, which needs the pump’s contact as well (sheet HW12)
Does the owner want the pump’s electricity measured, and is the pump single-phase or three-phase?	The pump meter: the SDM230 for a single-phase pump, the SDM630 for a three-phase one, or none (sheets HW06 and HW05)
What does the pump do when the tank is low and when it is full, and who is told today?	Nothing to buy. Greecon only reports: the float switches and trips already there stay the safety layer
Is there internet the box may use, where the enclosure will be?	The cellular router (sheet HW04)

The whole job at this kind of site, from the survey to the handover, is in the site pack for a water tank and pump station (PDF 07).

HW01

The box

The box is the small computer at the site. About every ten seconds it asks each device for its reading. It sends the readings to Greecon over the internet, keeps them while the internet is down, and says every 30 seconds that it is alive. It is also what looks for the equipment when a person presses **Find equipment**. It runs one small program and needs very little: what is hard is coming back after a power cut, and that is what its specification is about.

At a glance

WHAT IT DOES	Reads the meters and sensors and reports to Greecon. If the internet is down it keeps about three days of readings (for a site of up to about thirteen devices) and sends them in order when the internet returns.
HOW IT QUALIFIES	**Qualified by specification.** Any small computer that meets the specification on this sheet. What it must do well is come back by itself after a power cut.
SPECIFICATION	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input
FITTED BY	The project manager: prepared and tested at the office, plugged in on site.
PLANNING ESTIMATE (EUR)	180 to 300. Not a quote.
USED IN	Kits H, S, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

Any make qualifies that meets every line below. Nothing about a box is special, so a failed one can be replaced by anyone, with any box that qualifies.

It qualifies if:

- ☐ It runs Linux on a processor of the x86-64, 64-bit ARM or 32-bit ARM kind: Debian 12, Ubuntu 22.04 or later, or Raspberry Pi OS (64-bit), with systemd. The installer checks this.
- ☐ It has 4 GB of memory or more and 32 GB of storage or more. That is the least the software runs on; the kit list buys 8 GB and 128 GB.
- ☐ Its storage is a real disk: an SSD or eMMC. Not an SD card.
- ☐ It has a real-time clock with a battery, and a BIOS setting of “power on after power loss” (often “Restore on AC Power Loss: Power On”).
- ☐ It takes 9 to 36 V DC, or a plain 12 V, so that it can sit behind a DC UPS.
- ☐ It has no fan, and works from 0 to 50 °C or better: a plant room is hot and dusty.

- [ ] It has one Ethernet port. Two is nicer.

Option	When	Planning estimate (EUR)	Note
Fanless x86 mini PC (an Intel N100-class machine)	The default. Most sites	180 to 300	Check the clock battery and the power-on setting in the BIOS before buying many
Raspberry Pi 5 (4 or 8 GB)	A bench box, or a small site, with a proper SSD on NVMe or USB 3 and a clock battery. Never an SD card	120 to 200 with SSD and case	Starts by itself when power returns
Industrial PC (Advantech UNO, OnLogic, Seeed reTerminal DM class)	Hot, dusty or vibrating sites; customers who insist on industrial parts	450 to 900	The same software runs on it. Not needed to begin

STOP

Do not buy a box that starts from an SD card: it is the commonest reason these boxes fail, because an SD card corrupts on a power cut and wears out under constant writing. Do not buy one that has no clock battery, or that stays off when the power returns. Shop listings rarely say whether there is a clock battery: ask the vendor, in writing.

At each kind of site

Every site has one box. What changes is where it sits and what it reads.

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1	In the plant room. It reads the grid meter, the solar meter, the water module and the pump contact module: four devices, seven readings.
S Solar plant with grid connection	IN THE KIT	1	Near the main board or the inverter. It reads the grid meter and the solar meter and, where they are fitted, the load meter and the battery.
F Farm or greenhouse with irrigation	IN THE KIT	1	In the pump house. It reads the soil probes, the weather sensor, the water module and, if there is a pump to watch, the pump contact module. A 4G router is the likelier way to the internet.
W Water tank and pump station	IN THE KIT	1	In the pump house or the plant room. It reads the water module, the pump contact module and, if one is fitted, the pump meter.

Wherever it sits, the place is dry, has 30 cm free around the box, is away from heat, and is within reach of a socket that stays on and of a network cable.

What it does by itself

- It reads every device about every ten seconds and sends what it reads.

- It keeps about three days of readings if the internet is down, for a site of up to about thirteen devices (a site with more devices keeps fewer days), and sends them in order when the internet returns. In a long outage it keeps one reading a minute for each reading, so that three days fit.
- It reports every 30 seconds, so that Greecon knows it is alive. Greecon's own team is told when a customer's box has been silent for a quarter of an hour.
- It takes new versions of its software from the platform, and goes back to the version before if a new one fails to start three times.
- It restarts if it fails.
- It only reads. It never writes to a meter or a module. Nothing on the internet can connect to it: it only makes outgoing connections.

Settings, and where they are made

Everything is set at the office, on a desk, before the kit leaves. It takes about twenty minutes, then the install line.

Setting	What to do
The operating system	Debian 12 from the "netinst" image, with only "SSH server" and "standard system utilities". No desktop. One user with a long password, kept in the password manager against the site's name. Host name `greecon-<site>`
The clock	Time zone Europe/Tirane, and `timedatectl set-ntp true`
Security updates	`apt-get install -y unattended-upgrades`, and answer yes when it asks
The BIOS	Power on after power loss; no sleep; start from the disk only
The Greecon software	Paste the install line from the site's setup page (see *In the platform*)

What the installer changes, so that nobody wonders: `/opt/greecon` (the software), `/etc/greecon` (the box's credential, readable by the service only), `/var/lib/greecon-edge` (what it remembers), one service called `greecon-edge`, one user called `greecon`, and one command, `greecon-edge`. Nothing else.

Fitting

PROJECT MANAGER

- [ ] **Place:** in the enclosure if there is room, otherwise on a shelf beside it.
- [ ] **Power:** behind the UPS. A box that takes 9 to 36 V can share the 24 V supply; a box that takes 12 V keeps its own supply.
- [ ] **Network:** a cable to the customer's router or switch, or to the 4G router. Never a guest network. The box needs only outgoing HTTPS (port 443) to `api.greecon.earth` (and to `nodejs.org` when it is installed) and a time service (UDP 123). Nothing needs to come in.
- [ ] **Label** it with the site name and the date. Never write a code or a password on the box.

A box that was installed at the office keeps its credential when it is moved: unplug it, carry it to the site and plug it in, and it reports from the new place within a minute, with nothing typed again.

In the platform

The box is step 3 of the setup list, **Install the box**.

1. On the site's setup page press **Add the box and get its install command**. A line appears that begins `curl -fsSL https://api...` and ends in a code like `GRC-XXXX-XXXX-XXXX`. It is shown once. The code works once, for seven days. Copy it and press **I have copied it**.
2. Open a terminal on the box (a keyboard and screen, or SSH) and paste the line exactly. It runs in seven numbered steps and finishes with `Done. Greecon can see this box.`
3. Refresh the setup page. The step says **Done**, with the box's name, its version and "Heard from N seconds ago". Nobody pressed anything to make it so: the platform saw the box.

STOP

The install line is the box's key. Anyone with the line can install a box as this site until it is used or has expired. Treat it like a password: not in email or chat, and pasted straight into the box. If it is lost, or the seven days have passed, press **Install a box, or replace this one** for a new one.

To replace a box (failed, stolen or moved): press **Install a box, or replace this one** and confirm. A new line is shown, once. Install the operating system on the new box and paste the line. The new box takes the old one's place, and the old one's credential stops working at once. Nothing else is entered again: the addresses and the registers belong to the site, not to the box.

Acceptance test

Before a box leaves the office:

- ☐ The install line ran to `Done. Greecon can see this box.`
- ☐ The setup page shows the step done, with the box's name and "Heard from N seconds ago".
- ☐ `greecon-edge status` says when the platform last heard from the box, and does not say NOT REPORTING.
- ☐ The power was pulled and restored: the box started by itself.
- ☐ The box is labelled with the site name and the date.

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
Power cut. Unplug the box for 30 seconds, plug it in.	Back reporting within 3 minutes, and the box says nothing about readings lost			☐ Pass ☐ Fail
Internet cut (the first install, and any time the box or its software is new). Cut only the site's internet for ten minutes: unplug the router's uplink, or take the SIM out of the 4G router. Leave the box and the converter plugged into the router.	`greecon-edge status` shows a queue while it is out; the queue is empty a few minutes after the internet returns, and the readings from the gap are in the downloads			☐ Pass ☐ Fail

The commands a project manager uses on the box:

Command	What it does
`greecon-edge status`	How the box is: heard by Greecon, each device, the queue
`journalctl -u greecon-edge -n 50 --no-pager`	What the box has been saying
`greecon-edge find`	Looks at the network and says what answers. It only reads
`sudo systemctl restart greecon-edge`	Restarts the box's software

What goes wrong

What you see	Likely cause	What to do
The install line stops with `STOP:`	The sentence says which: no internet, a firewall that blocks outgoing HTTPS on port 443, a clock so far out that secure connections fail, or a code already used or older than seven days	Do what it says. If it stopped before step 5, *Giving this box its credential*, run the same line again. From step 5 on the code is used up: run the line again with no code at the end, which repairs the box and keeps its credential
The box says Offline	A power cut at the site (the commonest cause, and it ends by itself), the site's internet, or the box	Phone the site: is the power on, is the router's light on? Then, on the box: `greecon-edge status` and `journalctl -u greecon-edge -n 50 --no-pager`. The last lines say what is wrong
The box's log says the platform does not recognise this box's credential	The box was replaced, or its code was used for another box	Press **Install a box, or replace this one** and run the new line on the box
The setup page says its clock is N minutes fast	The box has no time service	`sudo timedatectl set-ntp true`
Readings are missing after a long outage	The queue holds about three days. It is for outages, not for a site that is never connected	Nothing is lost if the internet returns within that time. A site with no internet at all is not sold

More is in the runbooks (Project Manager Field Manual, PDF 02): R1 (a box is offline) and R10 (a box is replaced).

Not yet proven

- The heartbeat, the three-day queue and the updates with their way back are tested in software. They have not yet been shown on a real box with the internet really cut: that is why the internet-cut test is done at the first install, and its result written down.
- Installing the operating system and pasting the install line are still done in a terminal. There is not yet a box that sets itself up the first time it is switched on.
- The prices are planning figures from shop listings seen in October 2026, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	

HW02

The converter (RS-485 to Ethernet)

The meters, the modules and the field sensors speak Modbus RTU on a two-wire cable (RS-485). The box speaks Modbus TCP on the network. The converter sits between them: it puts the cable on the network, so that the box can reach every device on it. One of its settings matters more than all the others, and that is its mode.

At a glance

WHAT IT DOES	Puts the meters' cable (RS-485) on the network, so that the box can reach every meter and module on it. It must be in Modbus TCP to RTU gateway mode, and is set at the office (9600 baud, 8 data bits, no parity, 1 stop bit, TCP port 502).
HOW IT QUALIFIES	**Qualified by specification.** Any RS-485 to Ethernet converter that has a Modbus TCP to RTU gateway mode and takes the kit's serial settings and port 502.
SPECIFICATION	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode
FITTED BY	The project manager: set at the office, mounted and connected on site.
PLANNING ESTIMATE (EUR)	30 to 60 (basic) or 250 and up (industrial). Not a quote.
USED IN	Kits H, S, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

Any make qualifies that meets every line below. The box keeps one connection to a converter and asks one thing at a time, so any converter that does this will do.

It qualifies if:

- ☐ It converts RS-485 to Ethernet, fits a DIN rail and takes 9 to 36 V DC, so that it shares the 24 V supply.
- ☐ It has a **Modbus TCP to RTU gateway mode**. Makers also call it "Modbus gateway", "Modbus TCP server" or "protocol conversion".
- ☐ Its serial side can be set to 9600 baud, 8 data bits, no parity, 1 stop bit.
- ☐ Its TCP port can be set to 502.
- ☐ It can keep a fixed address on the network: set in the converter, or reserved for it in the router.

A basic converter costs about EUR 30 to 60 and an industrial one EUR 250 and up. Either qualifies.

STOP

Do not buy a converter that has only a “transparent” or “raw TCP” mode, or only “Modbus RTU over TCP”. In those modes the box’s requests arrive on the cable as noise, and nothing answers.

At each kind of site

Every site has one converter. What changes is what is behind it. The number after each device is its number on the cable.

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1	In the enclosure. Behind it, on one cable: the grid meter (number 1), the solar meter (2), the water module (3) and the pump contact module (5).
S Solar plant with grid connection	IN THE KIT	1	In the enclosure. Behind it: the grid meter (1), the solar meter (2) and, if fitted, the load meter (4). A battery that speaks Modbus on its own network address does not need the converter; one on RS-485 joins the cable with a number of its own.
F Farm or greenhouse with irrigation	IN THE KIT	1	In the enclosure in the pump house. Behind it: the water module (3), the pump contact module (5) if there is a pump to watch, the soil probe (6; a second probe takes 8) and the weather sensor (7).
W Water tank and pump station	IN THE KIT	1	In the enclosure. Behind it: the water module (3), the pump contact module (5) and, if fitted, the pump meter (4).

Settings, and where they are made

Greecon sets the converter in its own web page at the office, before the kit leaves. On site you check its label. The settings are here in case a converter ever has to be set again.

Setting	Value
Mode	Modbus TCP to RTU (gateway)
Serial: speed	9600 baud, the same as every device on the cable
Serial: data bits, parity, stop bits	8, none, 1
TCP: port	502
Address on the network	A fixed one: a reservation for the converter in the customer’s router (on its MAC address), or an address set in the converter itself, on the same range as the router. Arranged on site, and written down

Write the settings on the converter’s label and photograph its settings page.

Fitting

PROJECT MANAGER

- [ ] Mount it on the DIN rail in the enclosure, on the low-voltage side, and power it from the 24 V supply.
- [ ] Connect the data cable: A to A, B to B, and the ground where there is one, on to the first device and from there to the next. The rules of the cable are on sheet HW03.
- [ ] Plug its network socket into the same router or switch as the box (or into the 4G router). Never a guest network.
- [ ] Check its label against this sheet: gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit, port 502.
- [ ] Ask the customer's network person for a fixed address for it, and write the address down. You do not type it anywhere: the box finds the converter.

In the platform

The converter is not a device in the platform. It is the address of every device behind it, shown on each as *Reached at 192.168.1.50:502, unit 3*.

- **Finding it.** In step 4 of the setup list, **Where each device is on the network**, press **Find equipment**. The box tries port 502 on the private addresses of its own network (such as 192.168.x.x and 10.x.x.x), and behind whatever answers it asks unit numbers 1 to 10 a few read-only questions. It never writes to anything, and it takes about a minute.
- **If the converter is on another part of the network, or uses another port:** open **I know the converter's address**, type its address (each part without a leading zero: 192.168.1.50) and the port only if it is not 502, and press **Look at this address**. Only private addresses are accepted.
- **If its address changes later** (the router gave it a new one), every device behind it stops answering at the same moment. Press **Find equipment** again: the devices say **Found** at the new address, and **Use this address** sets them. Then ask for the fixed address.

Acceptance test

On a desk at the office, with the box:

- [ ] The converter's settings page shows gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit and port 502, and its label says the same.
- [ ] With one numbered device on its cable, the box finds the converter and that device at its own number:

```
greecon-edge find --host 192.168.1.50
```

It prints the converter's address and, under it, one line for each unit that answers, for example `unit 1: an Eastron meter, reading 1.80 kW at 231 V`.

- [ ] With two devices on the cable, each answers at its own number, with its own reading. If every number answers with the same words, the converter is not in gateway mode.

- [ ] A number that no device has does not answer.

On the day: Find equipment finds every device of the kit, and step 6, **Every device answers the box**, turns **Done** by itself.

What goes wrong

What you see	Likely cause	What to do
No Modbus converter answered on this network	The converter is off, or not on the same network as the box: a guest network, a second router, a cable in the wrong socket	Look at its power and network lights. Put the box and the converter on the same router. If it is on another part of the network, give its address
A converter answers at 192.168.1.50, but nothing answers behind it	The network side works and the cable side does not: A and B swapped, the converter not in gateway mode or not at 9600 baud, devices without power, a cable that is broken or branched	Swap the two wires at the converter. Check the mode and the speed in its settings page. Check that the devices have power
Units 1, 2 at 192.168.1.50 all answered with exactly the same words	The converter answers every unit number as the one device behind it: it is not in gateway mode	Set its mode to Modbus TCP to RTU (gateway), then press **Look again**
Every device behind it stops at the same moment (*no answer from 192.168.1.50:502 (unit 1)*), with the box online	The router gave the converter a new address	Press **Find equipment**, then **Use this address**. Ask for a fixed address, so that it does not happen again

More is in the runbooks (Project Manager Field Manual, PDF 02): R15 (a device is not found), R16 (a device is found at another address) and R17 (the converter's address changed).

Not yet proven

- This software has not yet looked at a real converter. It is tested against a pretend one that answers as the makers' documentation says. The first real kit is rehearsed on a desk before any site.
- Converters differ. Asked for a unit that is not there, some say so at once and some say nothing: the box handles both. One that accepts a single connection may drop the box's reading connection for a moment while the box looks, which costs one round of readings.
- A single device behind a converter in the wrong mode looks like a device at whichever number it is asked at: the box can tell the mode is wrong only when two units give the same answer. That is why the mode is checked on the settings page, and the first reading is compared with the meter's own display.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	

HW03

The data cable and labels

Every meter, module and field sensor of a kit hangs on one two-wire data cable (RS-485) that starts at the converter. Almost every fault that looks random comes from how this cable was run, not from what was bought, so this sheet is mostly rules. The labels are on the same sheet because they are fitted at the same time, and because the next person to open the enclosure has nothing else to go by.

At a glance

WHAT IT DOES	The RS-485 data cable (shielded twisted pair), network patch cords, ferrules and labels.
HOW IT QUALIFIES	**Qualified by specification.** Shielded twisted pair for the RS-485 line, ordinary network patch cords, ferrules and labels. How it is run matters more than what is bought.
SPECIFICATION	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels
FITTED BY	The project manager runs and labels it. The electrician connects its ends at any device that sits inside a distribution board.
PLANNING ESTIMATE (EUR)	40 to 80. Not a quote.
USED IN	Kits H, S, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

It qualifies if:

- ☐ The data cable is **shielded twisted pair**: one pair for A and B, with a conductor for the ground where the devices have a GND terminal.
- ☐ The network cables are ordinary patch cords, one for the box and one for the converter, tested with a cable tester.
- ☐ There are ferrules for every stranded end, and labels for every device and for both ends of every wire.

STOP

Do not use unshielded cable for the data line, do not run it in the same duct as power cables, and do not join devices with branches. A cable that qualifies and is run as a star still fails.

The rules of the cable

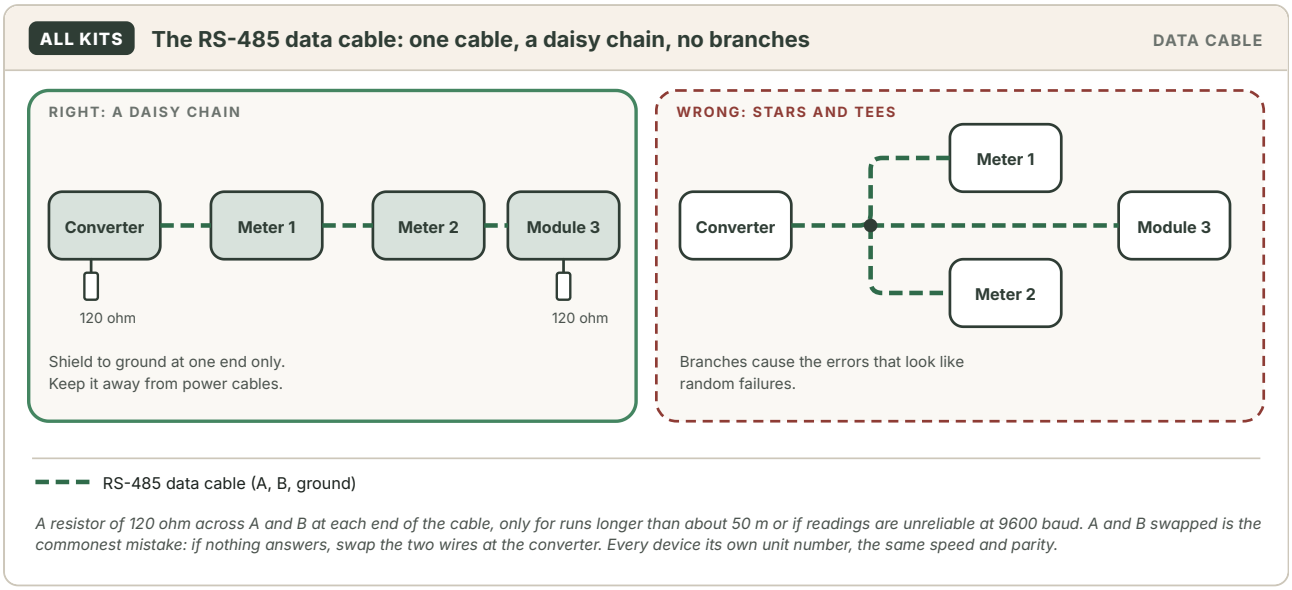


Figure 4-1. The RS-485 data cable: one cable and a daisy chain, never branches.

- **One cable, a daisy chain:** from the converter to the first device, on to the next, and so on. No branches: stars and tees cause the errors that look like random failures.
- **A to A and B to B at every device.** A and B swapped is the commonest mistake. If nothing answers, swap the two wires at the converter.
- **The shield joined to ground at one end only.** Keep the cable away from power cables, and note any place where it has to cross one.
- **Every device its own number, the same speed.** They are set at the office: on site you check each label and do not change it.
- **A 120 ohm resistor** across A and B at each end of the cable only for a run longer than about 50 metres, or if readings are unreliable at 9600 baud.
- **Plan the whole run under 100 metres.** A sensor further away than that needs another plan: ask the founder before promising it.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1 lot	One run from the converter through the grid meter and the solar meter (in the board, where the electrician connects it) and the water module to the pump contact module.
S Solar plant with grid connection	IN THE KIT	1 lot	One run from the converter through the grid meter, the solar meter and, if fitted, the load meter. Every end at a meter is the electrician's.
F Farm or greenhouse with irrigation	IN THE KIT	1 lot	The longest run of the four: from the converter in the pump house through the soil probe and the weather sensor in the field and the water module to the pump contact module. Plan it under 100 metres.
W Water tank and pump station	IN THE KIT	1 lot	A short run inside the pump house: from the converter through the pump meter (if fitted) and the water module to the pump contact module.

Labels

Label, on the day, and photograph before the enclosure is closed:

- [ ] The unit number on every device (it was set at the office), and the settings on the converter.
- [ ] The number of every wire, at both ends.
- [ ] The site name and the date, on a sticker inside the lid of the enclosure, and on the box.

Never write a code or a password on anything.

Fitting

PROJECT MANAGER runs the cable, and connects it at the converter, the modules and the field sensors. ELECTRICIAN connects its ends at any device inside a distribution board (the meters), because their terminals are inside the board.

- [ ] Walk the route first, and write down its length.
- [ ] Strip, fit ferrules, and connect A, B and the ground in the same order at every device.
- [ ] Join the shield to ground at the converter's end only.
- [ ] Fix the cable so that it cannot be pulled, and seal the glands.

In the platform

The cable is not in the platform. What it carries is: when it is right, **Find equipment** finds every device at its own number, and step 6 of the setup list, **Every device answers the box**, turns **Done** by itself.

Acceptance test

- [ ] The patch cords pass the cable tester.
- [ ] **Find equipment** finds every device of the kit, each at its own number.
- [ ] On the box, `greecon-edge status` lists every device as **ok**, and still does an hour later.
- [ ] The length of the run, and whether a resistor was fitted, are written on the commissioning record.

What goes wrong

What you see	Likely cause	What to do
Nothing behind the converter answers	A and B swapped	Swap the two wires at the converter
One device does not answer, the others do	That device's own A and B, its power, or the number on its label	Check its two wires and its power. Read its label against the install card. Numbers are not changed on site
Devices answer, then fail now and then	A branch in the cable, the shield joined to ground at both ends, a run beside power cables, or a long run with no resistor	Make it one line. Ground the shield at one end. Move it away from power. Over about 50 metres, fit the 120 ohm resistors
Two devices answer over each other	Two devices with the same number: one was not numbered at the office	That part goes back to the office, or the spare that was numbered there is fitted

Not yet proven

- The figures on this sheet (under 100 metres, a resistor over about 50 metres) are rules of thumb from the hardware guide, not measurements at a Greecon site. Write down the length of every run and whether a resistor was needed: the first sites will say whether the rule holds.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	

HW04

The cellular router

The box needs the internet, and it needs nothing else from it than to call out. Where the customer has a router the box may use, that is the best way and nothing is bought. Where there is none, as in a field or a pump station away from the buildings, a 4G router with a data SIM is the site's way to the internet, and the box and the converter plug into it.

At a glance

WHAT IT DOES	Gives the box and the converter an internet connection where the site has no usable one: a 4G router with a data SIM.
HOW IT QUALIFIES	**Qualified by specification.** Any 4G router for a DIN rail that takes a data SIM and has a network socket each for the box and the converter (or a small switch behind it).
SPECIFICATION	4G, DIN rail, data SIM
FITTED BY	The project manager.
PLANNING ESTIMATE (EUR)	150 to 200. Not a quote.
USED IN	Kit F; optional in kits H, S and W. The table under *At each kind of site* says what it does at each.

How it qualifies

It qualifies if:

- ☐ It is a 4G router that fits a DIN rail and takes a data SIM.
- ☐ It has a network socket for the box and one for the converter, or a small switch is fitted behind it.
- ☐ The supply it needs is known before it shares the 24 V supply: read its label.
- ☐ The SIM is a data SIM on a plan of 5 GB a month or more. Greecon supplies the SIM, charged at cost.

CHECK

Check the signal where the router will sit, at the survey: look at your phone there. A router in a steel enclosure in a pump house hears less than a phone in your hand.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	OPTIONAL	0 or 1	Only where the building has no internet the box may use. Most hotels have a router: then the box and the converter plug into that.
S Solar plant with grid connection	OPTIONAL	0 or 1	Only without usable internet. The hardware guide's own list for this kit does not show one; the network rules are the same as for every kit.
F Farm or greenhouse with irrigation	IN THE KIT	0 or 1	The likelier case: a field or a pump house seldom has a cable to the internet. The box and the converter plug into the router's sockets.
W Water tank and pump station	OPTIONAL	0 or 1	Only without usable internet: a pump station that stands away from the buildings.

A site with no internet at all is not sold. The box keeps about three days of readings, which is for outages, not for a site that is never connected.

How much data it uses

Plan for about **1.5 GB a month**. That assumes a hotel-size site (four devices, seven readings) sending one request every ten seconds, about 5 KB each with its headers (about 1.3 GB), and a report every 30 seconds (about 0.15 GB). A site with more devices sends more. It is an estimate: read the router's counter after a week and write the real figure down.

Fitting

PROJECT MANAGER

- [ ] Put the SIM in, and mount the router on the rail in the enclosure.
- [ ] Plug the box and the converter into its sockets.
- [ ] Check the signal on the router's own lights or page, where it sits, with the enclosure closed.
- [ ] Label it with the site name. Keep the SIM's number and its plan in the hardware register, not on the router.

In the platform

The router is not in the platform. What shows it is working is the box: step 3 of the setup list, **Install the box**, says **Done**, and the box stays **Online**.

Acceptance test

- [ ] With the box and the converter plugged into the router, the box comes **Online** within a minute.
- [ ] The internet-cut test of commissioning is done by taking the SIM out for ten minutes: `greecon-edge status` shows a queue, and the queue is empty a few minutes after the SIM is back.
- [ ] After a week, the router's data counter is read and written down, and the plan is checked against it.

What goes wrong

What you see	Likely cause	What to do
The box is Offline, and the power is on	No signal where the router sits, the plan's data is used up, or the SIM is not a data SIM	Look at the router's signal lights. Read its counter against the plan. A spare SIM is in the spares kit
The box goes Offline and comes back, again and again	A weak signal	Move the router's aerial outside the enclosure, as its manual allows. Readings are not lost: the box sends them when it is back
The data runs out before the month does	A site with more devices than the estimate assumes	Read the counter, and take a larger plan

More is in the runbooks (Project Manager Field Manual, PDF 02): R1 (a box is offline).

Not yet proven

- The 1.5 GB a month is worked out, not measured. No Greecon site has yet run on a cellular router.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	

HW05

Three-phase meter: Eastron SDM630

The Eastron SDM630 is the three-phase meter of every kit that measures electricity. Fitted where the supply enters the site, it gives both what is bought from the grid and what is sent to it. Fitted on the inverter's output, it gives what the solar plant produces. Greecon knows its registers, so nobody types one, and the box recognises it on the network.

At a glance

WHAT IT DOES	Measures electrical power on three phases. One at the grid connection (unit 1) gives both Grid import and Grid export; one on the inverter's AC output (unit 2) gives Solar production; a third, if wanted, on the site's own consumption (unit 4). Fitted by the electrician.
HOW IT QUALIFIES	**Approved model.** The Eastron SDM630 with Modbus. Greecon reads it by its own register table and recognises it on the network, so another three-phase meter does not take its place.
MAKE AND MODEL	Eastron SDM630 Modbus. The MCT version comes with three clamp-on current sensors for up to 100 A a phase; for more than 100 A, the ordinary SDM630 with 5 A current transformers
NUMBER ON THE CABLE	1 as the grid meter, 2 as the solar meter, 4 as the load meter. Set on its own display, at the office.
FITTED BY	The electrician, inside the distribution board, under the customer's lock-out. The project manager sets its number at the office and runs the data cable to the board.
PLANNING ESTIMATE (EUR)	90 to 180 each. Not a quote.
USED IN	Kits H and S. The table under *At each kind of site* says what it does at each.

How it qualifies

It is an approved model: the make and the model are named, and no other three-phase meter takes its place.

It qualifies if:

- ☐ It is an Eastron **SDM630** with Modbus (RS-485). The model on its label is the model on the quote.
- ☐ Its version suits the current. Up to 100 A a phase: the **MCT** version, which comes with three clamp-on current sensors. Above 100 A: the ordinary SDM630 with 5 A current transformers, bought to suit the supply.

☐ The maker's manual and Modbus protocol document came with it. Versions differ: read the one in its box.

STOP **Do not buy** another make of three-phase meter because it "also speaks Modbus". The box would not recognise it and the template's registers would not fit it. It could only be entered by hand, from its own register table, as equipment Greecon does not know.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	2	Two. The grid meter (number 1), where the supply enters, gives Grid import and Grid export. The solar meter (number 2), on the inverter's AC output, gives Solar production.
S Solar plant with grid connection	IN THE KIT	2, or 3	Two, as at a hotel: the grid meter (1) and the solar meter (2). A third (number 4) on the feed to the loads if the owner wants the site's own consumption measured and not worked out.
F Farm or greenhouse with irrigation	NOT USED	None	A farm kit has no energy meter. One can be added to any site from Add equipment Greecon already knows, and then this sheet applies.
W Water tank and pump station	NOT USED	None	The kit measures the pump's electricity with the single-phase SDM230 (sheet HW06). A three-phase pump needs this meter in its place: remove the template's pump meter and add Eastron SDM630 on the site's own consumption (number 4) from Add equipment Greecon already knows.

What Greecon reads from it

One number: the **total active power**. It is input register 52 (counted from zero; the manual's 30053), a 32-bit floating-point number, high half first, in watts. Greecon shows it in kW. The template fills the register in: nobody types it.

Fitted as	Reading in Greecon	How it is made from the power
Grid meter (number 1)	**Grid import**, kW	The power while it is positive: energy coming in
	Grid export, kW	The power while it is negative, shown as a positive number: energy sent out
Solar meter (number 2)	**Solar production**, kW	The size of the power, whichever way the clamps point
Load meter (number 4)	**Energy consumption**, kW	The size of the power

That is why one grid meter gives both import and export, and why a clamp fitted the wrong way round swaps them. Greecon notices a reversed grid meter on its own at night or in cloud, because energy cannot leave a site that nothing is producing for: the energy picture says so at once, and after five minutes a **Check today** alert opens, *The grid meter may be fitted the wrong way round*. At midday it cannot tell, which is why the direction check stays.

Settings, and where they are made

They differ between versions, so they are read from the meter's own display and never assumed. Greecon sets them at the office; on site you only check them, and photograph the display.

Setting	Value	Where
Unit number	1 on the grid meter, 2 on the solar meter, 4 on the load meter. A meter is delivered as 1, so each is numbered alone, before it joins the cable	On its display, at the office
Speed	9600 baud, which is how it is delivered: check it	On its display, at the office
Parity and stop bits	No parity, 1 stop bit	On its display, at the office
CT ratio	Only on a meter with 5 A current transformers: the ratio of the transformers fitted (for example 250 for a 250/5 A transformer), so that the meter shows the real power	In the meter's menu, on site, by the electrician

A meter's display works only when its voltage terminals have power, and powering a meter on a bench is mains work. The electrician makes up the office's bench lead. The project manager never wires mains.

Fitting

ELECTRICIAN Inside a distribution board, with the customer's lock-out on. The project manager watches, photographs and writes it down.

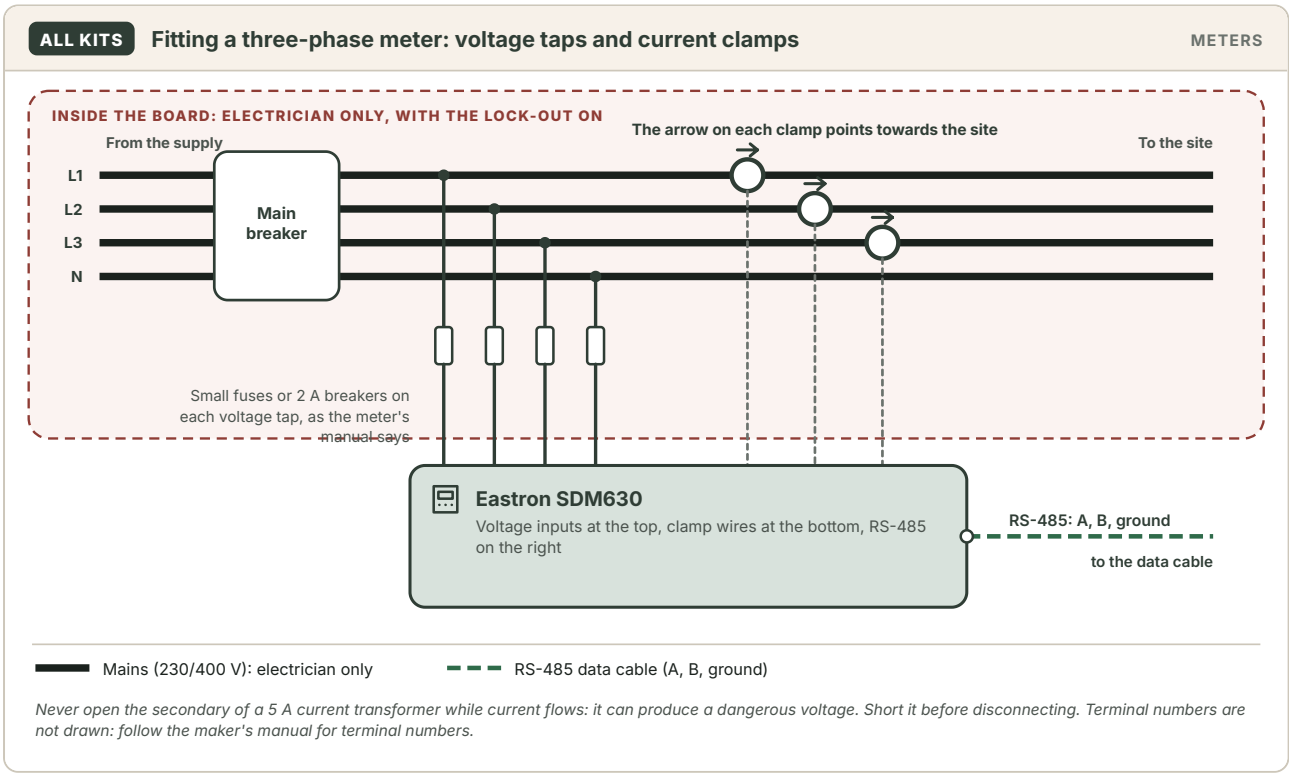


Figure 4-2. Fitting a three-phase meter: fused voltage taps and three current clamps, arrows towards the site. The electrician's work.

- [ ] **Clamps point towards the site.** On the grid meter this makes import positive. On the solar meter the clamps point away from the inverter, towards the board. Solar is read as the size of the power, so a reversed solar clamp does not read zero, but fit it the right way anyway.
- [ ] **Voltage taps** have small fuses or 2 A breakers on each tap, as the meter's manual says.
- [ ] **Never open the secondary of a 5 A current transformer while current flows:** it can produce a dangerous voltage. Short the secondary before disconnecting.
- [ ] The electrician connects the data cable at the meter, because its terminals are inside the board.
- [ ] **Photograph** the wiring, the label (model and serial number) and the display with its settings.

In the platform

- **Finding it.** In step 4 of the setup list, **Where each device is on the network**, press **Find equipment**. The box recognises an Eastron meter by what it says: a first-phase voltage that is a mains voltage, and a power. Found at the number the kit gives it, its address is filled in and the line says **Set**, for example *unit 1: an Eastron meter, reading 1.80 kW at 231 V*.
- **Found, but waiting for you.** A meter at another number than the kit's, or at another address than the one it has, says **Found** and is set only when you press **Use this address**.
- **A single-phase meter where this one was planned.** The line says *It is a single-phase SDM230, not the three-phase SDM630 planned.* If that is what the site has, press **Use this address** and Greecon reads it as an SDM230 (sheet HW06). If a three-phase meter was meant, do not press it: the electrician changes the meter.
- **The direction check** (step 9, **Grid meter direction checked**, at sites with a grid meter): with the sun down or the solar switched off, switch on a known load. Grid import must go up. Then press **I checked this**.

Acceptance test

At the office:

- [ ] The number and the speed are read from the meter's display, and photographed.
- [ ] On the desk, behind the converter, the box finds it at its number and reads its power:

```
greecon-edge find --host 192.168.1.50
greecon-edge modbus-test --host 192.168.1.50 --unit 1 --type input --address 52 --data-type
float32
```

The second prints the raw registers and the value, in watts.

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
The meter's display against Greecon. Read the display (kW or W) and Greecon at the same time.	Within 5 percent, or 1 kW on a small load			☐ Pass ☐ Fail
Grid direction. A known load on (a kettle or a heater), the sun down or the solar off.	Import rises; export does not			☐ Pass ☐ Fail
Solar at night.	Zero, or a few watts			☐ Pass ☐ Fail
Solar at midday. Against the inverter's own display.	Within about 5 percent			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
Import and export are swapped (export rises when a load is switched on)	The grid meter's clamps are the wrong way round	The electrician turns them, with the lock-out on. Repeat the direction check
The solar meter reads zero at midday	A clamp closed on the wrong cable, a breaker open upstream, or an inverter limited to zero export	Compare with the meter's own display. Ask the electrician to confirm that the clamps are on the inverter's output phases. On a zero-export plant, midday solar can honestly be low: check the inverter's display
Nothing that looks like an Eastron meter answers at unit 2	No power at the meter, the number or the speed on its display is not the kit's, or its A and B wires	Read its display against the install card. Check its two data wires. A meter powered from the grid is dark in a power cut: that is expected
Something answers … but not clearly enough to tell whether it is the three-phase SDM630 planned	One of its answers was lost on the cable	Press **Look again**. If it is the same, check the speed and the A and B wires
The reading is far from the display on a meter with current transformers	The CT ratio is not set in the meter	The electrician sets the ratio in the meter's menu. Greecon reads what the meter shows

More is in the runbooks (Project Manager Field Manual, PDF 02): R2 (a device does not answer), R3 (a reading looks wrong), R15 and R16 (Find equipment).

Not yet proven

- This software has not yet read a real SDM630. The register is from the maker's Modbus protocol document, and everything is tested against a pretend meter that answers as that document says. The first real kit is rehearsed on a desk, and the first reading at a site is compared with the meter's own display.

- The prices are planning figures from shop listings seen in October 2026, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	

HW06

Single-phase meter: Eastron SDM230

The Eastron SDM230 is the single-phase meter. In the standard kits it has one job: measuring the electricity of the pump at a water station. It is also the meter for a site whose whole supply is single-phase, where it takes the place of the three-phase SDM630. Greecon reads it by its own register, which is not the SDM630's, so it matters which of the two is fitted.

At a glance

WHAT IT DOES	Measures the electricity of one pump or a small load (single-phase), on its own feed (unit 4).
HOW IT QUALIFIES	**Approved model.** The Eastron SDM230 with Modbus. Greecon reads it by its own register table, which is not the SDM630's.
MAKE AND MODEL	Eastron SDM230, on the pump's feed, to measure its electricity
NUMBER ON THE CABLE	4 as the pump meter. Set on its own display, at the office, with its speed (9600).
FITTED BY	The electrician, on the pump's feed, under the customer's lock-out. The project manager sets its number and its speed at the office.
PLANNING ESTIMATE (EUR)	35 to 60. Not a quote.
USED IN	Optional in kit W. The table under *At each kind of site* says what it does at each.

How it qualifies

It is an approved model: the make and the model are named.

It qualifies if:

- ☐ It is an Eastron **SDM230** with Modbus (RS-485). The model on its label is the model on the quote.
- ☐ What it measures is single-phase: one pump, a small load, or a single-phase supply.
- ☐ The maker's manual came with it. Versions differ, and so does the speed a meter is delivered at: read the one in its box.

STOP

Fit the model the install card lists. An SDM630 fitted where an SDM230 was planned is read as an SDM230 and shows one phase's power only. A three-phase pump takes the SDM630 (sheet HW05), not this meter.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	NOT USED	None	Not in the standard kit, which is three-phase. On a single-phase supply it takes the SDM630's place at the grid connection (number 1) and on the inverter's output (number 2). Find equipment says which model it found.
S Solar plant with grid connection	NOT USED	None	As at a hotel: only on a single-phase plant, in the SDM630's place, with the same numbers (grid 1, solar 2, load 4).
F Farm or greenhouse with irrigation	NOT USED	None	A farm kit has no energy meter.
W Water tank and pump station	OPTIONAL	0 or 1	On the pump's own feed, as the pump meter (number 4). It gives Energy consumption: the pump's electricity. For a single-phase pump; a three-phase pump takes the SDM630 (sheet HW05).

What Greecon reads from it

One number: the **total active power**. It is input register 12 (counted from zero; the manual's 30013), a 32-bit floating-point number, high half first, in watts. Greecon shows it in kW. The template fills the register in: nobody types it.

Fitted as	Reading in Greecon	How it is made from the power
Pump meter (number 4), at a water station	**Energy consumption**, kW	The size of the power: the pump's electricity
Grid meter (number 1), on a single-phase supply	**Grid import** and **Grid export**, kW	The power while it is positive, and while it is negative
Solar meter (number 2), on a single-phase plant	**Solar production**, kW	The size of the power

Settings, and where they are made

Read from the meter's own display, never assumed. Greecon sets them at the office; on site you only check them, and photograph the display.

Setting	Value	Where
Unit number	4 as the pump meter (1 as a grid meter, 2 as a solar meter). It is delivered as 1	On its display, at the office
Speed	**Set it to 9600 baud.** The speed it is delivered at depends on its version: the V1.6 manual says 2400	On its display, on the Baud Rate page, at the office
Parity and stop bits	No parity, 1 stop bit	On its display, at the office

A meter's display works only when it has power, and powering a meter on a bench is mains work. The electrician makes up the office's bench lead. The project manager never wires mains.

Fitting

ELECTRICIAN On the feed of the pump (or where the single-phase supply enters), with the customer's lock-out on, as the meter's manual draws it. The project manager watches, photographs and writes it down.

- ☐ At a water station the meter sits in the board that feeds the pump, not in Greecon's enclosure.
- ☐ The electrician connects the data cable at the meter.
- ☐ **Photograph** the wiring, the label (model and serial number) and the display with its number and its speed.

In the platform

- **Finding it.** In step 4 of the setup list, **Where each device is on the network**, press **Find equipment**. Found at number 4 as the SDM230 that was planned, its address is filled in and the line says **Set**.
- **Where an SDM630 was planned** (a hotel or a solar plant on a single-phase supply): the line says *It is a single-phase SDM230, not the three-phase SDM630 planned. If you use this address, Greecon reads it as a single-phase SDM230.* Press **Use this address** if that is what the site has. The change is written in the audit trail, and the line keeps saying which model Greecon reads.
- **The other way round.** If the line says *It also answers the three-phase meter's total-power register, so it may be a three-phase SDM630*, look at the label on the meter. If it is an SDM630, remove this device and add the three-phase meter from **Add equipment Greecon already knows**.
- **No pump meter at this site:** press **The site does not have this: remove it** under it.

Acceptance test

At the office:

- ☐ The number and the speed (9600) are read from the meter's display, and photographed.
- ☐ On the desk, behind the converter, the box finds it at its number and reads its power:

```
greecon-edge find --host 192.168.1.50
greecon-edge modbus-test --host 192.168.1.50 --unit 4 --type input --address 12 --data-type
float32
```

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
The meter's display against Greecon. Read the display and Greecon at the same time, with the pump running.	Within 5 percent, or 1 kW on a small load			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
The meter does not answer, and the modules on the same cable do	Its speed was left as delivered (2400, the V1.6 manual says) and not set to the cable's 9600; or its number was left at 1	Read its speed and its number on its display against the install card. If they differ, the meter was not set at the office: take it back there, or fit the spare that was set there. Do not change it on site
The reading is about a third of what the display shows	The meter is a three-phase SDM630, read as an SDM230: one phase only	Check the label. Remove the device and add the three-phase meter from **Add equipment Greecon already knows**
The reading is zero while the pump runs	The meter is on the wrong feed	The electrician confirms which circuit it is on

More is in the runbooks (Project Manager Field Manual, PDF 02): R2 (a device does not answer), R3 (a reading looks wrong) and R16 (a device is found as another model).

Not yet proven

- This software has not yet read a real SDM230. The register and the delivered speed are from the maker's manual (V1.6), and everything is tested against a pretend meter. The first real kit is rehearsed on a desk, and the first reading at a site is compared with the meter's own display.
- The prices are planning figures from shop listings seen in October 2026, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	

HW07

The battery (the customer's own)

A battery at a solar plant is the customer’s own equipment. Greecon buys nothing for it and approves no model. It can be read if, and only if, its maker publishes where its charge and its power are: its Modbus register table. Then its address is entered by hand and its two readings are mapped by hand. If the maker will not publish the table, the battery is left out. Nobody guesses a register.

At a glance

WHAT IT DOES	Reports the battery’s charge (percent) and its power (kW, positive while charging), from the battery maker’s own Modbus registers. It is the customer’s own equipment: Greecon buys nothing for it.
HOW IT QUALIFIES	**Qualified by its register table.** Only if the battery’s maker publishes its Modbus register table. Greecon never guesses a register.
SPECIFICATION	Whatever battery the customer has, provided its maker publishes its Modbus register table. It speaks Modbus on the site’s network, at an address of its own, or on its own RS-485 port
NUMBER ON THE CABLE	Its own, entered by hand (for example 8), if it joins the data cable. None if it is on the network at an address of its own.
FITTED BY	Nobody: it is already there. The project manager enters its address and maps its two readings.
PLANNING ESTIMATE (EUR)	Nothing to buy: it is the customer’s own.
USED IN	Optional in kit S. The table under *At each kind of site* says what it does at each.

How it qualifies

It qualifies if:

- ☐ Its maker publishes the battery’s **Modbus register table**, and you have it in your hand: the register of the state of charge and of the power, the kind of number each is, and its scale.
- ☐ It speaks Modbus where the box can reach it: on the site’s network at an address of its own (Modbus TCP), or on its own RS-485 port.
- ☐ On RS-485, it can be set to the cable’s speed (9600 baud, 8 data bits, no parity, 1 stop bit) and to a number no other device has.
- ☐ Reading it needs no password that changes, and no cloud account: the box reads registers and nothing else.

STOP

If the maker will not give the register table, leave the battery out, and say so at handover. A register taken from a forum, or from another model, would show a charge that looks right and is not. Greecon only reads: it never sets a battery's mode, its limits or anything else.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	NOT USED	None	The hotel template has no battery.
S Solar plant with grid connection	OPTIONAL	0 or 1	The template's Battery system (optional). Two readings are mapped by hand: Battery charge and Battery power. If the plant has no battery, remove the device.
F Farm or greenhouse with irrigation	NOT USED	None	None.
W Water tank and pump station	NOT USED	None	None.

What Greecon reads from it

Reading in Greecon	Unit	What it must be	Standard alert
Battery charge	%	The state of charge, 0 to 100	**Low below 25 percent**, a Check today alert. Set it above the reserve the battery system keeps for itself
Battery power	kW	The battery's power, **positive while charging** and negative while it discharges	None

A charge outside 0 to 100, or a power beyond 2,000 kW either way, is called a reading that cannot be real.

Settings, and where they are made

Nothing is set on the battery by Greecon. What is entered is in the platform, by the project manager, from the maker's table:

What	Where it comes from
Its address	The battery's own address on the network; or, on RS-485, the converter's address
Its port	The maker's table (502 is usual)
Its unit number	The maker's table; on the data cable, a number no other device has (for example 8)
For each reading: the register kind, the address, the kind of number, which half comes first, and the scale	The maker's table

Fitting

Nothing is fitted. The battery is already there, and its installer's work is not touched.

- ☐ If it joins Greecon's data cable (RS-485), PROJECT MANAGER connects A, B and the ground at the battery's communication port, as the battery's manual draws it, at the end of the daisy chain. Nothing inside the battery's power side is opened.
- ☐ If it is on the network, nothing is connected: the box reaches it through the router.

In the platform

The solar template makes the device **Battery system (optional)** with its two readings and no registers.

1. **No battery at this plant:** press **The site does not have this: remove it** under it in step 4, and nothing waits for it.
2. **Its address** (step 4, **Where each device is on the network**): the box cannot recognise a battery on a network, so its line says **Enter by hand**. Under the device open **Enter the address by hand**, fill in the **IP address**, the **Port** and the **Unit number**, and press **Save**.
3. **Its readings** (step 5, **Which register holds which reading**): under each reading press **Choose the register**. The register kind is **Input** for a manual's 3xxxx and **Holding** for 4xxxx. The address counts from zero (40001 is address 0). Choose the kind of number and, for a number of two registers, which half comes first. Under how it becomes the reading choose **Multiply and add**: a battery that reports watts for a reading in kW is multiplied by 0.001. Press **Save the register**.
4. **Its limit** (step 10, **Alert limits agreed with the customer**): agree the low-charge limit with the owner.

Acceptance test

☐ Each register is read from the box before it is relied on, and the number is the one on the battery's own display:

```
greecon-edge modbus-test --host 192.168.1.60 --unit 8 --type holding --address 0 --data-type uint16 --scale 0.1
```

(The address, the unit, the kind and the scale are the maker's: these are an example.)

Check and how	Pass when	Greecon shows	Real	Result
Battery charge. Against the battery's own display.	The two agree (write both numbers)			☐ Pass ☐ Fail
Battery power. While the battery is charging, and again while it discharges.	Positive while charging, negative while discharging, and the size agrees with the battery's display			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
A reading is a thousand times too big	Watts shown as kilowatts	In the register form, multiply by 0.001
The power has the wrong sign (negative while charging)	The maker counts the other way	In the register form, under *Multiply by*, enter a negative number (for example -0.001), so that charging is positive
A reading is constant, or zero	A wrong register, or the wrong register kind (input against holding)	Read the register from the box twice, a minute apart, against the maker's table
The battery does not answer	Its address, its port or its unit number; on RS-485, its speed or A and B	Check each against the maker's table and the battery's own settings page

Not yet proven

- No battery has been read by this software, and no battery model is approved. Every battery is a first time: the acceptance test above is the proof, for that battery.
- The energy picture shows a battery only from these two readings. If only one can be mapped, say at handover which one is missing.

What to write down

There is nothing to buy. Write on the commissioning record:

Field	Answer
Battery make, model and firmware version	
Where the register table came from (document and date)	
Address, port and unit number entered	
Register, kind of number and scale of the charge	
Register, kind of number and scale of the power	
Acceptance test passed on (date), by	

HW08

4 to 20 mA module: Waveshare Analog Input 8CH

The level, pressure and flow transmitters each send a current between 4 and 20 mA down two wires. This module reads those currents and answers the box with numbers, on the same data cable as the meters. The platform calls it the **Water I/O module**. It is an approved model, so the box finds it, and the only thing a person is asked for is the range printed on each transmitter's label.

At a glance

WHAT IT DOES	Reads the 4 to 20 mA signals of the level, pressure and flow transmitters (inputs 1, 2 and 3) and answers the box on the same RS-485 cable as the meters, as unit 3. It is left in its default 0 to 20 mA mode, so that a wire that has come off reads zero and Greecon can say so.
HOW IT QUALIFIES	**Approved model.** The Waveshare Modbus RTU Analog Input 8CH, in the version that reads current. The version that reads voltage is a different part.
MAKE AND MODEL	Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC
NUMBER ON THE CABLE	3. Set with the maker's setting tool, at the office (it is delivered as 1).
FITTED BY	The project manager mounts it in the enclosure and joins it to the data cable. The electrician wires the 4 to 20 mA loops into its inputs.
PLANNING ESTIMATE (EUR)	30 to 60. Not a quote.
USED IN	Kits H, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

It is an approved model: the make and the model are named.

It qualifies if:

- ☐ It is the Waveshare **Modbus RTU Analog Input 8CH**, in its current version: eight inputs, on RS-485, powered from 7 to 36 V DC.
- ☐ It is the version that **reads current**. The version that reads voltage is a different part, and does not qualify.
- ☐ The maker's setting tool for it is on the office laptop, with a USB-to-RS-485 adapter: its number is set by a command, not by a switch.

STOP

Do not buy another make of analog input module, or the voltage version of this one. The box recognises this module by a register that holds its own number, and reads its inputs as microamps: another module answers differently, or not at all.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1	One. Input 1 takes the tank's level transmitter, input 2 the pipe's pressure transmitter, input 3 the flow meter if there is one. It gives Tank level, Line pressure and Pump flow.
S Solar plant with grid connection	NOT USED	None	A solar plant has no water side.
F Farm or greenhouse with irrigation	IN THE KIT	1	One, for the tank and the pipe behind the irrigation. The flow meter (input 3) only if there is a pump to watch.
W Water tank and pump station	IN THE KIT	1	One, the centre of the kit: level on input 1, pressure on input 2, the flow meter on input 3 if it is fitted.

What Greecon reads from it

Each input is one input register that holds the current in microamps: input 1 is register 0, input 2 is register 1, input 3 is register 2. A 4 to 20 mA transmitter reads 4000 to 20000. Greecon turns that into the reading from the range on the transmitter's label.

Input	Transmitter	Reading in Greecon	The label gives	Standard alert
1	Level (sheet HW10)	**Tank level**, %	The range, in metres or bar, and how tall the tank is	Critical below 15 % (Act now); Low below 35 % (Check today)
2	Pressure (sheet HW11)	**Line pressure**, bar	The top of the range, in bar	None
3	Flow (sheet HW12), if fitted	**Pump flow**, litres a minute	The top of the range, in litres a minute	No flow at or below 0.2 litres a minute (Act now), only when the pump's running state is read too

Its inputs are left in their default 0 to 20 mA mode, on purpose. A wire that has come off then reads 0, which is below the 4 mA of an empty tank, so Greecon can see it and say so (*cannot be a tank level*). In the module's own 4 to 20 mA mode a broken wire might read like an empty tank.

Settings, and where they are made

Setting	Value	Where
Its number	**3.** It is delivered as 1, and its number is kept in a register: there is no switch. Set it with the maker's setting tool and a USB-to-RS-485 adapter, with no other device on the cable	At the office
Speed	9600 baud, 8 data bits, no parity, 1 stop bit, which is how it is delivered. Check it	At the office
Input mode	Left at the default, 0 to 20 mA. Do not change it	Nowhere

Label the module with its number, and photograph the label.

Fitting

PROJECT MANAGER mounts the module on the DIN rail in the enclosure, powers it from the 24 V supply and joins it to the data cable. ELECTRICIAN wires the transmitters' loops into its inputs.

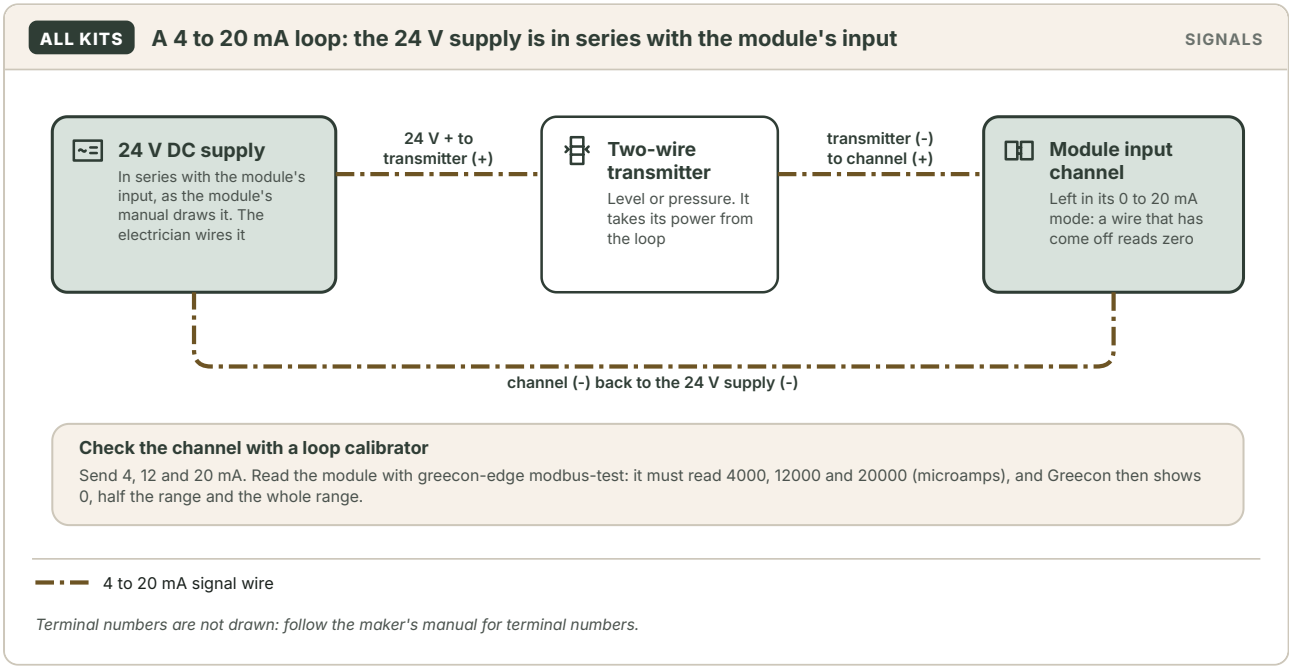


Figure 4-3. A 4 to 20 mA loop: the 24 V supply in series with the module's input.

- ☐ **Each transmitter has its own input:** level on input 1, pressure on input 2, the flow meter on input 3.
- ☐ **A two-wire transmitter takes its power from the loop,** so each loop has the 24 V supply in series with the module's input, wired as the module's manual draws it. The maker does not say that the module powers the loop, so the loop has a supply of its own.
- ☐ Number every wire at both ends, and photograph the terminals before the enclosure is closed.

In the platform

- **Finding it.** In step 4 of the setup list, **Where each device is on the network**, press **Find equipment**. The box recognises the module by the register that holds its own number, and by its inputs. Found at number 3, its address is filled in: *unit 3: a 4 to 20 mA input module, reading 12.0 mA, 9.0 mA, 5.1 mA*.
- **The ranges.** In step 5, **Which register holds which reading**, the page asks one thing for each transmitter, in the words of its label: the **Range on the label** and the **Unit on the label** (metres or bar) of the level transmitter and **How tall the tank is inside**; the **Top of the range** of the pressure transmitter, and of the flow meter. Press **Save the ranges**. Greecon works out the register and the scale: nobody types either.
- **A transmitter the site does not have** (no flow meter): tick **Not fitted at this site**. The page asks first, because the reading and its history are removed.

Acceptance test

At the office, on the desk:

- ☐ The module is numbered 3, alone on the cable, and labelled.
- ☐ Behind the converter, the box finds it at number 3 (`greecon-edge find --host 192.168.1.50`).
- ☐ With a loop calibrator in place of a transmitter (a hand-held that sends 4, 12 and 20 mA), the module reads 4000, 12000 and 20000 on that input. Address 0 is input 1, 1 is input 2, 2 is input 3:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 3 --type input --address 0
```

- ☐ With nothing connected to an input, it reads 0.

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
4 to 20 mA channels. With a loop calibrator, send 4, 12 and 20 mA into each input that is used.	The reading at the bottom, the middle and the top of its range			☐ Pass ☐ Fail

On the tank's input, the top of the range reads more than 100 when the transmitter is longer than the tank (125 for a 0 to 5 m transmitter in a tank 4 m tall), and Greecon may call that impossible while you test. That is the test, not a fault.

What goes wrong

What you see	Likely cause	What to do
A reading below zero (*-31.2 cannot be a tank level*)	No current in the loop: a wire has come off, or the loop has lost its 24 V supply. The module reads 0	Read the input from the box. 0 means no current. The electrician checks the loop and its supply

What you see	Likely cause	What to do
A reading far too high or too low	The range, or the tank's height, was given wrongly; or a label in bar was entered as metres (it then reads about ten times too low)	Read the label again and give the ranges again
The right number on the wrong reading	Two transmitters on each other's inputs	Level on input 1, pressure on input 2, flow on input 3
Nothing that looks like a 4 to 20 mA input module answers at unit 3	The module has no power, was left at number 1, or its A and B wires	Check its power and its two data wires. Read its label. A module that was not numbered at the office goes back there, or the spare that was numbered there is fitted

More is in the runbooks (Project Manager Field Manual, PDF 02): R15 (a device is not found) and R18 (a reading is impossible after the ranges were given).

Not yet proven

- **This software has not yet read a real one of these modules.** Its registers (the inputs at input registers 0 to 7, in microamps; its number at holding register 0x4000) are from the maker's published pages, read again on 31 October 2026, and everything is tested against a pretend module that answers as those pages say. The first real kit is rehearsed on a desk, and anything that differs from this sheet is written down and reported to Greecon.
- The prices are planning figures from shop listings seen in October 2026, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	

HW09

Pump contact module: Waveshare IO 8CH

This module tells Greecon whether the pump is running. A spare contact of the pump's starter closes when the pump runs; the module reads that contact and answers the box, on the same data cable as everything else. It only reports. The module also has relay outputs, and they are never connected: nothing Greecon fits can switch a pump.

At a glance

WHAT IT DOES	Reports whether the pump is running, from a spare contact of its starter (a dry contact on input 1), as unit 5. It only reports; it never switches anything.
HOW IT QUALIFIES	**Approved model.** The Waveshare Modbus RTU IO 8CH. Only its digital inputs are used: its relay outputs are never connected.
MAKE AND MODEL	Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used
NUMBER ON THE CABLE	5. Set with the maker's setting tool, at the office (it is delivered as 1).
FITTED BY	The project manager mounts it and joins it to the data cable. The electrician wires the pump starter's spare contact to its input 1.
PLANNING ESTIMATE (EUR)	30 to 60. Not a quote.
USED IN	Kits H and W; optional in kit F. The table under *At each kind of site* says what it does at each.

How it qualifies

It is an approved model: the make and the model are named.

It qualifies if:

- ☐ It is the Waveshare **Modbus RTU IO 8CH**: eight isolated digital inputs, on RS-485. A dry contact on an input is enough.
- ☐ The maker's setting tool for it is on the office laptop, with a USB-to-RS-485 adapter: its number is set by a command, not by a switch.
- ☐ The pump's starter has a **spare auxiliary contact** that follows the pump (closed while it runs). This is asked at the survey. Without one, the module is not fitted.

STOP

Never connect anything to the module's relay outputs, and never wire its input to a control circuit. The contact only reports. Greecon only reads, and it is not a safety system: the float switches and trips already in the customer's equipment stay the safety layer.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1	One: it reads a spare running contact of the water pump's starter and gives Pump running. Leave it out if the starter has no spare contact, and say so at handover.
S Solar plant with grid connection	NOT USED	None	A solar plant has no pump to watch.
F Farm or greenhouse with irrigation	OPTIONAL	0 or 1	Only if there is a pump to watch and its starter has a spare running contact.
W Water tank and pump station	IN THE KIT	1	One, for the station's pump. Leave it out if the starter has no spare running contact.

What is lost without it. With no running contact, Greecon cannot tell a pump that has stopped from one that is running dry, and the alert "the pump is running but no water is flowing" cannot be raised. The site then removes the module from its list, and the handover says so.

What Greecon reads from it

Input	Reading in Greecon	What it means	Standard alert
1	**Pump running**	1 while the pump runs, 0 while it is stopped	None of its own. With a flow meter (sheet HW12) it makes the no-flow alert possible: pump running, flow at or below 0.2 litres a minute for 30 seconds, an Act now alert

Input 1 is the module's first digital input (read as a discrete input, address 0). The template fills it in: nobody types it. Inputs 2 to 8 are not used.

Settings, and where they are made

Setting	Value	Where
Its number	**5.** It is delivered as 1, and its number is kept in a register: there is no switch. Set it with the maker's setting tool and a USB-to-RS-485 adapter, with no other device on the cable	At the office
Speed	9600 baud, 8 data bits, no parity, 1 stop bit, which is how it is delivered. Check it	At the office

Label the module with its number, and photograph the label.

Fitting

PROJECT MANAGER mounts the module on the DIN rail in the enclosure, powers it from the 24 V supply and joins it to the data cable. ELECTRICIAN wires the starter's contact to it.

- [ ] A **spare auxiliary contact** of the pump's starter goes to **input 1** as a dry contact, as the module's manual draws it. The module's DI COM is left unconnected.
- [ ] Nothing is connected to the relay outputs.
- [ ] Number the two wires at both ends, and photograph the terminals before the enclosure is closed.

In the platform

The templates call it **Pump contact module (optional)**.

- **Finding it.** In step 4 of the setup list, **Where each device is on the network**, press **Find equipment**. The box recognises the module by the register that holds its own number, and by its inputs. Found at number 5, its address is filled in: *unit 5: a digital input module, input 1 is closed*.
- **No spare contact at this site:** press **The site does not have this: remove it** under it. The device and its reading go, and nothing waits for them.
- **Nothing else is typed.** Its reading is mapped by the template.

Acceptance test

At the office, on the desk:

- [ ] The module is numbered 5, alone on the cable, and labelled.
- [ ] Behind the converter, the box finds it at number 5 (`greecon-edge find --host 192.168.1.50`).
- [ ] With a wire across input 1 as the manual draws a dry contact, it reads 1; with the wire off, 0:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 5 --type discrete --address 0
```

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
Pump running. The owner's own person starts the pump, then stops it.	Greecon shows running within 30 seconds, and changes both ways			☐ Pass ☐ Fail

Do not start or stop the pump yourself: the owner's own person does, because they know what else it affects.

What goes wrong

What you see	Likely cause	What to do
The pump shows running when it is stopped (and, with a flow meter, the no-flow alert opens while the pump is stopped)	The contact is inverted: it is closed when the pump is stopped	The electrician uses the other contact of the starter, or swaps its normally-open and normally-closed wires. Until it is right, the no-flow alert cannot be trusted
The reading never changes	The contact does not follow the pump (it is a contact of something else), or a wire is off	The electrician confirms which contact it is. Read the input from the box while the pump is started
Nothing that looks like a digital input module answers at unit 5	The module has no power, was left at number 1, or its A and B wires	Check its power and its two data wires. Read its label. A module that was not numbered at the office goes back there

More is in the runbooks (Project Manager Field Manual, PDF 02): R3 (a reading looks wrong) and R15 (a device is not found).

Not yet proven

- **This software has not yet read a real one of these modules.** Its registers (the digital inputs read with function 02 at addresses 0 to 7; its number at holding register 0x4000) are from the maker's published pages, read again on 31 October 2026. The maker's page warns that its example frames do not always match their labels. Everything is tested against a pretend module that answers as the pages say. The first real kit is rehearsed on a desk, and anything that differs from this sheet is reported to Greecon.
- The planning price (EUR 30 to 60) is the earlier figure for a pump run input. It has not been checked against a shop price for this module.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	

HW10

Level transmitter

The level transmitter says how full the tank is. It hangs at the bottom of the tank, where the weight of the water above it becomes a current between 4 and 20 mA on two wires: 4 mA when the tank is empty, more as it fills. The 4 to 20 mA module (sheet HW08) reads that current, and Greecon turns it into a percentage of the tank from two things a person gives: the range printed on the transmitter's label, and how tall the tank is inside.

At a glance

WHAT IT DOES	Measures how full the tank is. It hangs at the bottom of the tank.
HOW IT QUALIFIES	**Qualified by specification.** Any submersible 4 to 20 mA level transmitter whose range reaches the top of the tank and whose wetted parts suit the water it hangs in.
SPECIFICATION	Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)
WIRED TO	Input 1 of the 4 to 20 mA module (sheet HW08), in a loop with the 24 V supply.
FITTED BY	The electrician or the customer's plumber fits it and the electrician wires its loop. Nobody from Greecon enters a tank.
PLANNING ESTIMATE (EUR)	120 to 400. Not a quote.
USED IN	Kits H, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

Any make qualifies that meets every line below. The tank decides what is bought: how tall it is, what it holds, and how a cable gets out of it. All three are asked at the survey.

It qualifies if:

- ☐ It is a submersible level transmitter with a vented cable and a two-wire 4 to 20 mA output.
- ☐ **Its range is at least the tank's height.** A 0 to 5 m transmitter suits a tank up to 5 m tall. One shorter than the tank cannot see the top of the water, and Greecon refuses it.
- ☐ **Its wetted parts suit the water.** Into a drinking-water tank goes only a transmitter whose wetted parts are approved for drinking water, with the maker's document that says so.
- ☐ Its cable reaches a dry place above the water with no joint under water.
- ☐ Its label gives its range, in metres or in bar. Photograph the label before the transmitter goes into the tank: afterwards nobody can read it.

Kind	When	What to know
Submersible transmitter, hung at the bottom of the tank	The default	Greecon asks for the range on its label and for the tank's height
Non-contact ultrasonic sensor, above the water	Where nothing may hang in the water, or where no document shows that a submersible one is approved for drinking water	It must have a 4 to 20 mA output that is 4 mA when the tank is empty and rises as the tank fills. That is set in the sensor's own menu, from its manual. See *Not yet proven*

STOP

Do not put a sensor of unknown material into a drinking-water tank. A cheap sensor is not worth a contaminated tank. If the maker cannot show that the wetted parts are approved for drinking water, use a non-contact ultrasonic sensor above the water.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1	One, at the bottom of the hotel's water tank. Only a transmitter with wetted parts approved for drinking water goes into a drinking-water tank. It gives Tank level.
S Solar plant with grid connection	NOT USED	None	A solar plant has no water side.
F Farm or greenhouse with irrigation	IN THE KIT	1	One, in the irrigation tank or reservoir. It gives Tank level, with the standard low-level alerts.
W Water tank and pump station	IN THE KIT	1	One, in the station's tank. If the tank holds drinking water, the drinking-water rule applies.

What Greecon reads from it

One reading: **Tank level**, as a percentage of the tank. 0 is empty and 100 is full.

- The module reads the loop's current on its input 1, in microamps: 4000 at 4 mA and 20000 at 20 mA.
- 4 mA is 0 percent. 20 mA is the top of the transmitter's range, as a share of the tank: the range ÷ the tank's height × 100.
- A label in bar is turned into metres of water: one bar holds up 10.197 m.

The label says	The tank is	A full tank reads	Half full reads	20 mA would read
0 to 5 m	4 m tall	100 %, at 16.8 mA	50 %, at 10.4 mA	125 %
0 to 5 m	5 m tall	100 %, at 20 mA	50 %, at 12 mA	100 %

The label says	The tank is	A full tank reads	Half full reads	20 mA would read
0 to 0.5 bar (5.1 m of water)	4 m tall	100 %, at 16.6 mA	50 %, at 10.3 mA	127 %

Standard alert	Opens when	Kind
Critical	The level is below 15 %	Act now
Low	The level is below 35 %	Check today: plan a refill

The two limits are the standard ones. The site's own are agreed with the owner. A level a little under 0 or over 100 (up to 5 points) is still taken as real: a transmitter has a tolerance, and a tank is measured a few centimetres short. Beyond that Greecon says that the number cannot be a tank level.

Settings, and where they are made

Nothing is set on a submersible transmitter: its range is fixed when it is made. What Greecon needs is read and measured:

What	Value	Where it comes from
The range on the label	For example 5, for a 0 to 5 m transmitter, and its unit: metres or bar	The label, read before the transmitter goes into the tank
How tall the tank is inside	In metres, from the bottom, where the transmitter hangs, to the top of the water when the tank is full	Measured at the survey. If the tank must be opened, the customer's own staff measure it
Its input on the module	Input 1	Wired by the electrician

Fitting

ELECTRICIAN or the customer's plumber fits the transmitter, and the electrician wires its loop. **PROJECT MANAGER** stays outside the tank, photographs and writes down.

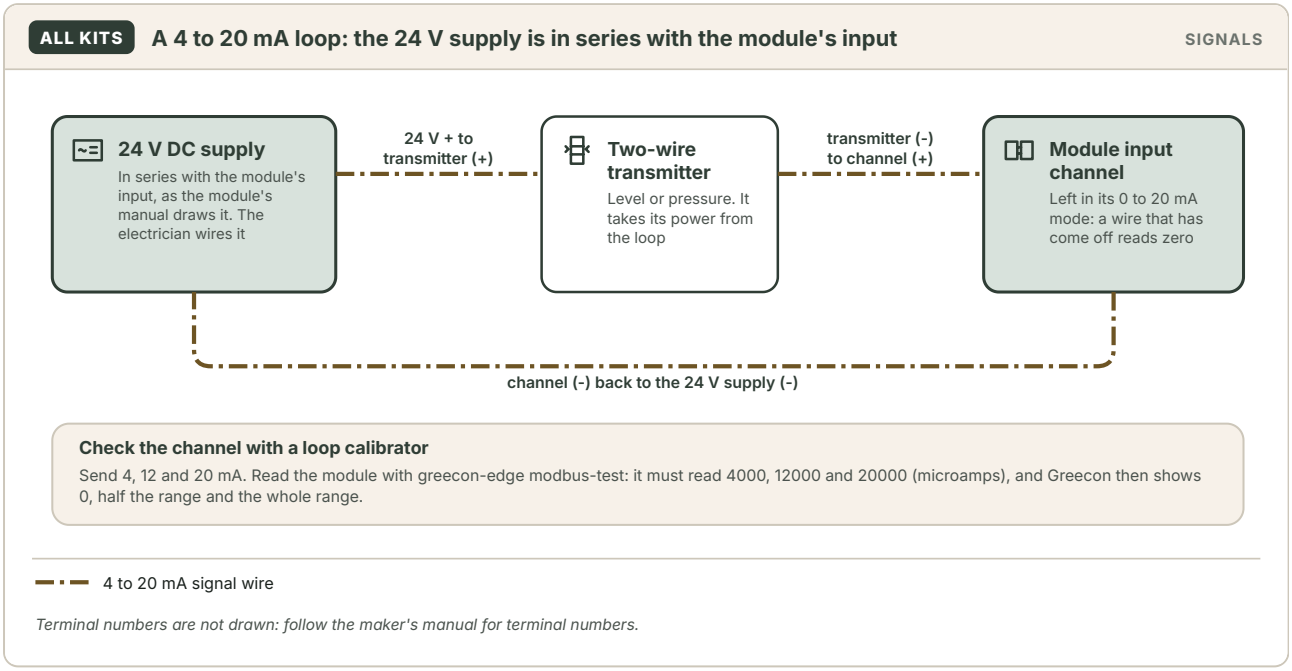


Figure 4-3. A 4 to 20 mA loop: the 24 V supply in series with the module's input.

- ☐ The transmitter hangs **at the bottom of the tank,** so that zero is empty, and its cable is fixed so that it cannot move.
- ☐ The **vented cable** ends in a dry place. The small tube inside it must stay open to the air and must not take in water: it is not kinked, sealed or left where it can get wet.
- ☐ The loop is wired to **input 1** of the module, with the 24 V supply in series, as the module's manual draws it.
- ☐ The two wires are numbered at both ends, and the terminals are photographed.

STOP **Nobody from Greecon enters a tank.** If the tank must be entered, the customer's own staff do that. If the only place for the sensor is inside and nobody may go in, say that it is not possible, and write why.

In the platform

The transmitter is not a device in the platform. It is one reading of the **Water I/O module**, which the box finds in step 4, **Where each device is on the network**.

- **Its range.** In step 5, **Which register holds which reading**, under **Level transmitter**, give the **Range on the label** (5), the **Unit on the label** (metres or bar) and **How tall the tank is inside** (4), and press **Save the ranges**. The page says what it set: *Set now: 4 mA is 0 % and 20 mA is 125 %.* Greecon works out the register and the scale: nobody types either.
- **A number is read the way people write it:** "2,5" is 2.5.

- **A transmitter shorter than the tank is refused** when you save: *it cannot see the top of the water. Fit a transmitter with a longer range.*
- **To change the range later,** type both numbers again: the form does not show the old ones.
- **The limits.** In step 10, **Alert limits agreed with the customer**, the standard ones are in place (critical below 15, low below 35). Agree the site's own with the owner.

Acceptance test

At the office, before the kit leaves:

- ☐ The label is photographed, and its range and its unit are written on the install card.
- ☐ With a loop calibrator on input 1 in place of the transmitter, the module reads 4000, 12000 and 20000 (sheet HW08).
- ☐ In its loop on the desk, in the air, the transmitter reads about 4000: 4 mA, an empty tank.

```
greecon-edge modbus-test --host 192.168.1.50 --unit 3 --type input --address 0
```

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
Tank level. Against a dipstick or the sight gauge, at two levels. Take the second after the tank has been used or filled.	Within 5 percentage points, both times			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
A level below zero (*-31.2 cannot be a tank level*)	No current in the loop: a wire has come off, or the loop has lost its 24 V supply. The module reads 0	Read input 1 from the box: 0 means no current. The electrician checks the loop and its supply
A level far above 100 (*133 cannot be a tank level*)	The range given is bigger, or the tank's height smaller, than the real one. A tank 4 m tall given as 3 m, with a 0 to 5 m transmitter, reads 133 when it is full	Read the label again, measure the tank again, and give both numbers again
A level about ten times too low	A label in bar was entered as metres	Choose bar under **Unit on the label**, and save the ranges again
The level is steady, but off by more than 5 points	The height was measured to the wrong place, or the transmitter does not hang at the bottom	Measure from where the transmitter hangs to the top of the water when the tank is full

What you see	Likely cause	What to do
The level drifts with the weather	The tube in the vented cable is blocked, kinked or wet, so the transmitter no longer compares with the open air	The electrician or the plumber frees the end of the cable and keeps it dry
The level jumps about	The transmitter swings in the water, near the inlet	Fix the cable so that the transmitter cannot move

More is in the runbooks (Project Manager Field Manual, PDF 02): R3 (a reading looks wrong) and R18 (a reading is impossible after the ranges were given).

Not yet proven

- This software has not yet read a level transmitter through a real 4 to 20 mA module. The scale is tested against a pretend module. The first real one is checked against a dipstick at two levels, and both numbers are written down.
- **No ultrasonic sensor has been tried.** The form's words are written for a transmitter that hangs in the water. For a sensor above the water, set it in its own menu to send 4 mA when the tank is empty and 20 mA at a level you choose, and give that level as the range (20 mA at 4 m, in a tank 4 m tall: range 4, height 4). Report what you find to Greecon.
- The prices are planning figures from shop listings seen in October 2026, not quotes. Level transmitters were seen from about USD 28, of unverified quality, to over GBP 900.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	

HW11

Pressure transmitter

The pressure transmitter says what the pressure in the pipe is. It is screwed into the pipe behind an isolating valve, so that it can be taken out without draining the pipe, and it sends a current between 4 and 20 mA on two wires: 4 mA at no pressure and 20 mA at the top of its range. The 4 to 20 mA module (sheet HW08) reads that current on its input 2, and Greecon asks for one number: the top of the range on the transmitter's label.

At a glance

WHAT IT DOES	Measures the pressure in the pipe, through an isolating valve so that it can be removed without draining.
HOW IT QUALIFIES	**Qualified by specification.** Any 4 to 20 mA pressure transmitter with stainless wetted parts and a range that suits the pipe.
SPECIFICATION	4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts
WIRED TO	Input 2 of the 4 to 20 mA module (sheet HW08), in a loop with the 24 V supply.
FITTED BY	A plumber makes the wet connection, through an isolating valve. The electrician wires its loop.
PLANNING ESTIMATE (EUR)	40 to 150. Not a quote.
USED IN	Kits H, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

Any make qualifies that meets every line below. The pipe decides what is bought: its pressure range is asked at the survey.

It qualifies if:

- ☐ It has a two-wire 4 to 20 mA output.
- ☐ **Its range suits the pipe.** The top of its range is above the highest pressure the pipe sees. 0 to 10 bar is the usual one.
- ☐ It measures gauge pressure: zero in the open air, like the pressure gauge on the line that it is checked against.
- ☐ Its wetted parts are stainless steel. On a drinking-water pipe, ask the plumber whether the transmitter and its fitting may be used there.
- ☐ Its thread is known before the visit, so that the plumber brings the right fitting and the isolating valve.

- [ ] Its label gives its range, in bar. Photograph the label.

> **STOP** **Do not fit it without an isolating valve.** Without one the transmitter cannot be taken out, checked or changed without draining the pipe. And do not buy the kind with a voltage output (0 to 10 V): the module reads current.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1	One, on the pipe the pump feeds. It gives Line pressure.
S Solar plant with grid connection	NOT USED	None	A solar plant has no water side.
F Farm or greenhouse with irrigation	IN THE KIT	1	One, on the irrigation line.
W Water tank and pump station	IN THE KIT	1	One, on the pipe the pump feeds. The pipe's pressure range, asked at the survey, decides the transmitter's range.

What Greecon reads from it

One reading: **Line pressure**, in bar. 4 mA is 0 bar, and 20 mA is the top of the range.

The label says	4 mA	12 mA	20 mA	With a wire off, Greecon shows
0 to 10 bar	0 bar	5 bar	10 bar	-2.5 bar, and says that it cannot be a line pressure
0 to 6 bar	0 bar	3 bar	6 bar	-1.5 bar, and says the same
0 to 4 bar	0 bar	2 bar	4 bar	-1.0 bar, which Greecon accepts as a pressure. See *What goes wrong*

A pressure outside -1 to 40 bar is called a reading that cannot be real. There is **no standard alert** on the pressure: Greecon shows it and its trend.

Settings, and where they are made

Nothing is set on the transmitter: its range is fixed when it is made.

What	Value	Where it comes from
The top of the range	In bar: 10, for a 0 to 10 bar transmitter	The label
Its input on the module	Input 2	Wired by the electrician

Fitting

A plumber makes the wet connection, and ELECTRICIAN wires the loop. PROJECT MANAGER watches, photographs and writes down. The pipe is the customer's: it is opened by the customer's plumber, or by one the customer agrees to.

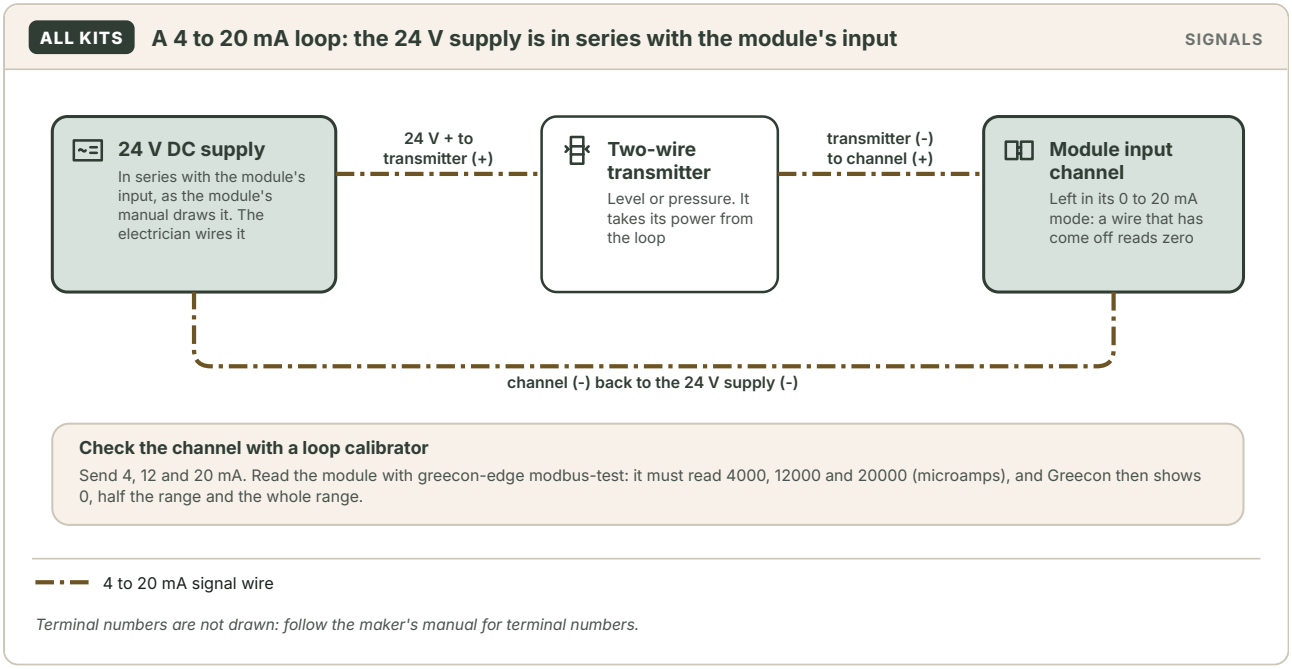


Figure 4-3. A 4 to 20 mA loop: the 24 V supply in series with the module's input.

- ☐ The transmitter goes on the pipe the pump feeds, **through an isolating valve.**
- ☐ Where there is a pressure gauge on the line already, the transmitter goes near it, so that the two can be compared.
- ☐ The loop is wired to **input 2** of the module, with the 24 V supply in series, as the module's manual draws it.
- ☐ The two wires are numbered at both ends. The label, the valve and the terminals are photographed.

In the platform

The transmitter is not a device in the platform. It is one reading of the **Water I/O module**, which the box finds in step 4, **Where each device is on the network**.

- **Its range.** In step 5, **Which register holds which reading**, under **Pressure transmitter**, give the **Top of the range** (10) and press **Save the ranges**. Greecon works out the register and the scale: nobody types either.
- **A number is read the way people write it:** "2,5" is 2.5.

Acceptance test

At the office, before the kit leaves:

- ☐ The label is photographed, and the top of its range is written on the install card.
- ☐ With a loop calibrator on input 2 in place of the transmitter, the module reads 4000, 12000 and 20000 (sheet HW08).
- ☐ In its loop on the desk, in the air, the transmitter reads about 4000: 4 mA, no pressure.

```
greecon-edge modbus-test --host 192.168.1.50 --unit 3 --type input --address 1
```

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
Pressure. Against the pressure gauge on the same line.	Within 0.2 bar			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
A pressure below zero that never moves (-2.5 with a 0 to 10 bar transmitter)	No current in the loop: a wire has come off, or the loop has lost its 24 V supply. With a transmitter of 0 to 4 bar or less the number is between -1 and 0, and Greecon does not call it impossible	Read input 2 from the box: 0 means no current. The electrician checks the loop and its supply
Greecon and the gauge agree at rest and differ under pressure	The top of the range was given wrongly (a 0 to 16 bar transmitter given as 10)	Read the label again and give the range again
Zero in Greecon while the gauge shows pressure	The isolating valve is closed	The plumber, or the customer's own person, opens it
The right number on the wrong reading	The transmitter is on another input	Pressure is on input 2

More is in the runbooks (Project Manager Field Manual, PDF 02): R3 (a reading looks wrong) and R18 (a reading is impossible after the ranges were given).

Not yet proven

- This software has not yet read a pressure transmitter through a real 4 to 20 mA module. The scale is tested against a pretend module. The first real one is checked against the gauge on the line, and both numbers are written down.

- The prices are planning figures from shop listings seen in October 2026, not quotes. A known make of pressure transmitter was seen from GBP 129.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	

HW12

Flow meter

The flow meter says how much water is moving in the pipe, in litres a minute. By itself it gives a number and its trend. Together with the pump's running contact (sheet HW09) it gives the alert a pump station most wants: the pump is running but no water is flowing. It is optional in every kit, and it is the dearest of the transmitters, so the survey decides whether the site gets one.

At a glance

WHAT IT DOES	Measures the flow in the pipe. Without it (or without the pump's running contact) the alert for a pump that runs with no water moving cannot be raised.
HOW IT QUALIFIES	**Qualified by specification.** Any flow meter for the pipe and the fluid that has a 4 to 20 mA output.
SPECIFICATION	4 to 20 mA output. Needed for "the pump is running but no water is flowing"
WIRED TO	Input 3 of the 4 to 20 mA module (sheet HW08). Its manual says whether its output powers the loop itself or needs the 24 V supply.
FITTED BY	A plumber fits it in the pipe, with the straight run its manual asks for. The electrician wires its signal.
PLANNING ESTIMATE (EUR)	150 to 500. Not a quote.
USED IN	Optional in kits H, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

Any make qualifies that meets every line below. An electromagnetic meter and a turbine meter are the usual kinds.

It qualifies if:

- ☐ It suits the pipe and the water: the pipe's size and material, and clean or dirty water. On a drinking-water pipe, its wetted parts are approved for drinking water.
- ☐ **It has a 4 to 20 mA output.** A meter with only a pulse output does not qualify: the module reads current.
- ☐ Its manual says which flow 20 mA stands for (the top of its range), or lets that be set, and that flow is above the most the pump delivers.
- ☐ Its manual says whether its output powers the loop itself, or needs the loop's 24 V supply. The electrician needs to know which.
- ☐ The pipe has the straight run the manual asks for, before and after the meter. Look at the survey.

STOP

Do not buy a flow meter that is not rated for the fluid, or one whose only output is pulses. Fitting it means cutting the pipe: that is a plumber's work, agreed with the customer, with the pump stopped by the customer's own person.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	OPTIONAL	0 or 1	With the pump contact module it makes the no-flow alert possible: the pump is running but no water is flowing.
S Solar plant with grid connection	NOT USED	None	A solar plant has no water side.
F Farm or greenhouse with irrigation	OPTIONAL	0 or 1	Only if there is a pump to watch.
W Water tank and pump station	OPTIONAL	0 or 1	The part behind a pump station's most useful alert: pump running, no flow. Without it the handover says that the alert cannot be raised.

What is lost without it. Without a flow meter, or without the pump's running contact, Greecon cannot say that the pump is running but no water is flowing. The site then ticks **Not fitted at this site**, and the handover says that this alert cannot be raised.

What Greecon reads from it

One reading: **Pump flow**, in litres a minute. 4 mA is no flow, and 20 mA is the top of the range.

The top of the range	4 mA	12 mA	20 mA
100 litres a minute (6 cubic metres an hour)	0	50	100
500 litres a minute (30 cubic metres an hour)	0	250	500

Greecon asks for litres a minute. A range in cubic metres an hour is multiplied by 16.67, and one in litres a second by 60.

Standard alert	Opens when	Kind
No flow	The pump is running (Pump running reads 1) and the flow has been at or below 0.2 litres a minute for 30 seconds	Act now

The alert needs the pump's running state to be read as well. With no pump contact module it is never raised. A flow below zero, or above 100,000 litres a minute, is called a reading that cannot be real.

Settings, and where they are made

What	Value	Where it comes from
The top of the range	The flow that 20 mA stands for, in litres a minute	The meter's label or its manual; on a meter with a menu, what is set there
Its input on the module	Input 3	Wired by the electrician
The no-flow limit	0.2 litres a minute as standard. See the check below	In the platform, with the owner

CHECK **Read the flow with the pump stopped and the pipe full.** A meter at rest seldom sends exactly 4 mA. If Greecon then shows 0.4 litres a minute, a limit of 0.2 would never open the alert. Agree a limit a little above the resting number and far below the pump's ordinary flow, and write both numbers down.

Fitting

A plumber fits the meter in the pipe, and ELECTRICIAN wires its signal. PROJECT MANAGER watches, photographs and writes down.

- ☐ The meter sits in the pipe the pump feeds, with the **straight run** its manual asks for before and after it, and the arrow on its body pointing the way the water goes.
- ☐ Its signal is wired to **input 3** of the module, as the meter's manual and the module's manual draw it.
- ☐ **One supply in the loop, never two.** A meter whose output powers the loop is wired without the 24 V supply; one that needs the loop's supply has the 24 V supply in series, as for the other transmitters (sheet HW08).
- ☐ A meter that needs mains power is connected by the electrician.
- ☐ The wires are numbered at both ends. The label and the terminals are photographed.

In the platform

The flow meter is not a device in the platform. It is one reading of the **Water I/O module**, which the box finds in step 4, **Where each device is on the network**.

- **Its range.** In step 5, **Which register holds which reading**, under **Flow meter**, give the **Top of the range** in litres a minute and press **Save the ranges**. Greecon works out the register and the scale: nobody types either.
- **No flow meter at this site:** tick **Not fitted at this site** under it. The page asks first, because the reading and its history are removed.
- **The limit.** In step 10, **Alert limits agreed with the customer**, the limit is called *No flow at or below*. The standard is 0.2 litres a minute.

Acceptance test

At the office, before the kit leaves:

- ☐ The label is photographed, and the top of its range, in litres a minute, is written on the install card.
- ☐ With a loop calibrator on input 3 in place of the meter, the module reads 4000, 12000 and 20000 (sheet HW08).

```
greecon-edge modbus-test --host 192.168.1.50 --unit 3 --type input --address 2
```

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
Flow. A bucket test (fill a measured volume and time it), or the flow meter's own display.	Within 10 percent			☐ Pass ☐ Fail
At rest. The pump stopped by the owner's own person, the pipe full.	The flow is below the no-flow limit. Write the number down			☐ Pass ☐ Fail

Do not start or stop the pump yourself: the owner's own person does, because they know what else it affects.

What goes wrong

What you see	Likely cause	What to do
A flow far below zero (*-125 cannot be a flow*)	No current in the loop: a wire has come off, the loop has no supply, or the meter has no power. The module reads 0	Read input 3 from the box: 0 means no current. The electrician checks the loop
A flow a little below zero while the pump is stopped (*-0.2 cannot be a flow*)	The meter rests a hair under 4 mA. The box reads about 4000, not 0	Check the reading with the pump running. Write the resting number down and report it to Greecon
The no-flow alert does not open when the pump runs dry	The meter's resting number is above the limit, or the pump's running state is not read	Set the limit above the resting number. Check that Pump running reads 1 while the pump runs (sheet HW09)
The no-flow alert opens while the pump is stopped	The pump's contact is inverted	The electrician uses the starter's other contact (sheet HW09)
The flow is a steady share too high or too low	The top of the range was given in the wrong unit (cubic metres an hour for litres a minute), or is not what the meter is set to	Read the meter's own setting, and give the range again
A flow below zero at a site that has no flow meter	The reading was left in place, and an empty input reads like a wire off	Tick **Not fitted at this site** under the flow meter

More is in the runbooks (Project Manager Field Manual, PDF 02): R3 (a reading looks wrong) and R18 (a reading is impossible after the ranges were given).

Not yet proven

- This software has not yet read a flow meter through a real 4 to 20 mA module. The scale and the no-flow alert are tested against pretend equipment.
- **How a real meter rests is not known.** The module reads in small steps, and a meter at rest may sit a little above or below 4 mA. That is why the resting number is read and written down at commissioning, and why the limit is agreed after it.
- The prices are planning figures, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	

HW13

Soil moisture probe

A soil moisture probe sits in the soil of an irrigation zone and says how wet the soil is at its depth. It speaks Modbus RTU on the same data cable as everything else. **No model of soil probe is approved yet.** Greecon does not know any probe's registers, and the box cannot recognise one on a network. So a probe qualifies by its maker's register table, its address is entered by hand, and its reading is mapped by hand from that table.

At a glance

WHAT IT DOES	Measures soil moisture at the probe's depth. Its depth and place are written down, because soil readings mean little without them. Unit 6, mapped by hand from the maker's register table.
HOW IT QUALIFIES	**Qualified by its register table.** A soil probe that speaks Modbus RTU on RS-485, whose maker publishes its register table, and whose number and speed can be set.
SPECIFICATION	RS-485 Modbus, with a published register table, one for each zone
NUMBER ON THE CABLE	6 (a second probe takes the next free number, 8). Set the maker's way, at the office, before the probe is buried.
FITTED BY	The project manager, with the grower, who says where and how deep.
PLANNING ESTIMATE (EUR)	40 to 120 each. Not a quote.
USED IN	Kit F. The table under *At each kind of site* says what it does at each.

How it qualifies

It qualifies if:

- ☐ It speaks **Modbus RTU on RS-485,** on two data wires.
- ☐ **Its maker publishes its register table,** and you have it in your hand before you buy: the register that holds the moisture, the kind of number it is, and its scale.
- ☐ Its unit number and its speed can be set, and the maker says how: to a number of its own, and to 9600 baud, 8 data bits, no parity, 1 stop bit.
- ☐ What it reports becomes a percentage. The reading in Greecon is **Soil moisture**, from 0 to 100, and the table says how the probe's number becomes one.
- ☐ It is made to stay in the soil, and its cable reaches a dry junction with no joint in the ground.
- ☐ Its supply voltage is one the kit has: read its label against the 24 V supply.

STOP

Do not buy a probe whose register table you have not seen. A probe whose maker does not document its registers cannot be mapped honestly, so it cannot be used. **And never bury a probe before its number is set:** one that was not must be dug up.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	NOT USED	None	None.
S Solar plant with grid connection	NOT USED	None	None.
F Farm or greenhouse with irrigation	IN THE KIT	1 for each zone	One for each irrigation zone, at the depth and in the place agreed with the grower. It gives Soil moisture. Greecon gives no irrigation advice until the grower's own Start irrigating below value is entered.
W Water tank and pump station	NOT USED	None	None.

What Greecon reads from it

One reading for each probe: **Soil moisture**, in percent.

What is real	0 to 100. Outside that, Greecon says that the number cannot be soil moisture
Standard alert	None
The limit	*Start irrigating below*. It has **no standard value**: the same percentage is waterlogged in sand and bone dry in clay, and it depends on the crop, the depth of the roots and the probe
Until the limit is entered	Greecon gives no irrigation advice. It shows the reading and its trend
Once it is entered	Greecon says when the soil has fallen below it. That is ordinary work on a farm, not an alert

Soil readings are only as good as the contact between the probe and the soil, and they mean little without the probe's depth and place. Both are written down.

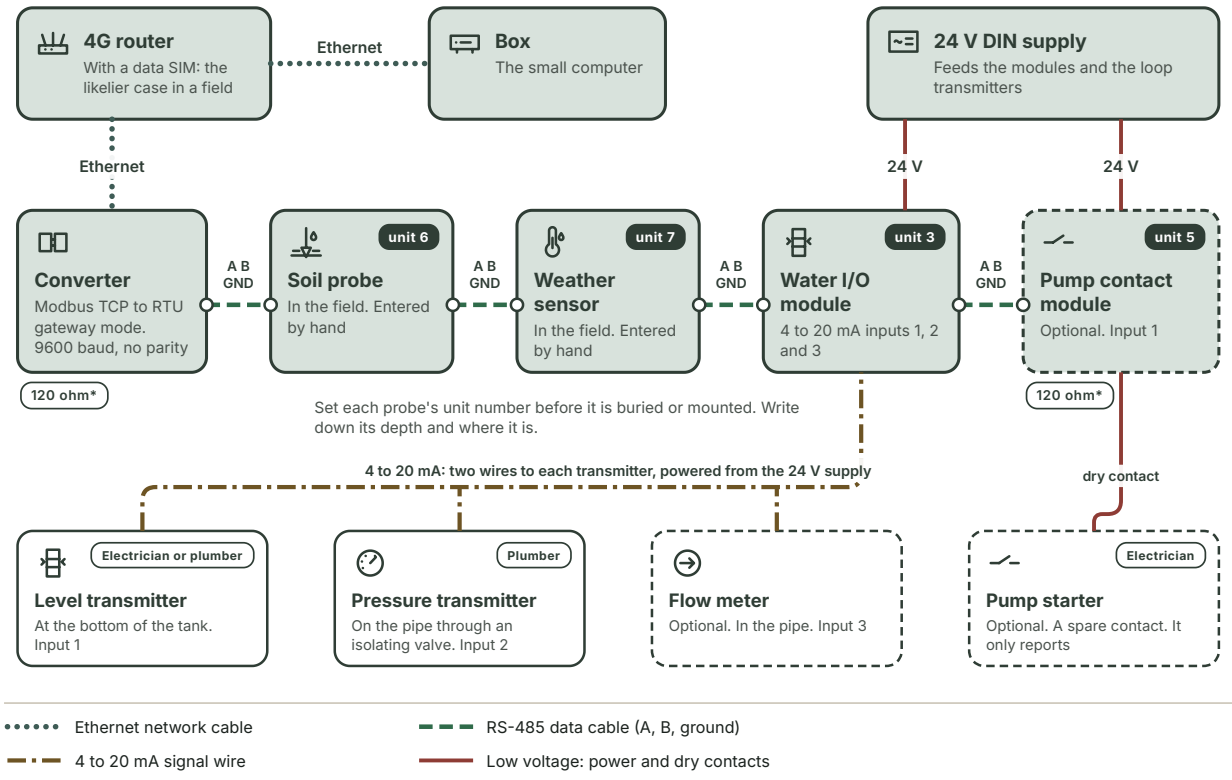
Settings, and where they are made

Setting	Value	Where
Unit number	**6.** A second probe, for a second zone, takes the next free number: 8, then 9. Many probes are delivered as 1, the same as another device. Set it the maker's way, with the USB-to-RS-485 adapter and no other device on the cable	At the office
Speed	9600 baud, 8 data bits, no parity, 1 stop bit, set the maker's way	At the office
Its label	The number, written on the probe's cable at the end that stays above the ground	At the office

Fitting

PROJECT MANAGER with the grower, who says where and how deep.

KIT F **Wiring: the data cable and the signal wires** BUS AND SIGNALS



**Only for runs longer than about 50 m, or if readings are unreliable at 9600 baud. The soil probe and the weather sensor need a published register table, and their addresses are entered by hand. The electrician wires the 4 to 20 mA loops and the contact. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.*

Figure F-2. Kit F. Wiring: one RS-485 cable from the converter through the soil probe, the weather sensor and the water module to the pump contact module, with the 4 to 20 mA loops into the water module and the pump starter's contact into the pump contact module.

☐ **The place is typical of the zone:** not its wettest spot and not its driest. Walk the field with the grower and choose it together.

- [ ] **The depth is where the roots are.** Ask the grower.
- [ ] The probe is placed as its maker says, **in good contact with the soil,** with no air gap around it.
- [ ] It joins the data cable in the daisy chain: A to A, B to B, no branches (sheet HW03). Plan the whole run under 100 metres.
- [ ] **Write down each probe's depth and place, and photograph it** before the soil is closed.

In the platform

The farm template makes the device **Soil probe** with its reading and no register.

1. **Its address** (step 4, **Where each device is on the network**). The box cannot recognise a probe, so its line says **Enter by hand**. Under the device open **Enter the address by hand** and fill in the **IP address** (the converter's: the water module's line shows it, as *Reached at 192.168.1.50:502, unit 3*), the **Port** (502) and the **Unit number** (6). Press **Save**.
2. **Its register** (step 5, **Which register holds which reading**). Under the reading press **Choose the register**. The register kind is **Input** for a manual's 3xxxx and **Holding** for 4xxxx. The address counts from zero (30001 is address 0). Choose the kind of number, and under how it becomes the reading choose **Multiply and add**: a probe that reports tenths of a percent is multiplied by 0.1. Press **Save the register**.
3. **A second zone.** On the site's **Equipment and readings** page, under *Add equipment or a reading by hand (advanced)*, add a second soil probe and its reading. Then give it its address and its register as for the first, with its own unit number (8).
4. **The limit** (step 10, **Alert limits agreed with the customer**). Leave *Start irrigating below* empty on the day. After the first week of readings, ask the grower which reading means "time to water", and enter it.

Write what you enter:

Probe	Unit number	Register kind	Address	Kind of number	Multiply by	Depth and place
Zone 1	6					
Zone 2	8					

Acceptance test

At the office, on the desk, before anything is buried:

- [ ] The probe has its number (6), alone on the cable, and is labelled.
- [ ] Behind the converter, its register is read from the box, and the number is believable:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 6 --type input --address 0 --data-type uint16
--scale 0.1
```

(The kind, the address, the kind of number and the scale are the maker's: these are an example.)

- [ ] In the air it reads low, and in a glass of water it reads high.

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
Soil probe. In the air, then in a glass of water, then in the soil.	Three different readings: low in the air, high in the water, plausible in the soil			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
The probe does not answer	A wrong unit number (left as delivered, the same as another device), a number or an address entered wrongly, a speed that is not the cable's, A and B swapped, or no power	Check each, in that order. Test the register from the box against the maker's table
The reading never changes, or is the same in the air and in water	A wrong register, the wrong register kind (Input against Holding), or a wrong scale	Read the maker's table again and test the register from the box
The reading is ten times too big or too small	The scale	Under **Multiply and add**, multiply by the table's factor
The reading is believable but does not follow watering	Poor contact with the soil, or a place that is not typical of the zone	Place the probe again, with the grower. Write the new depth and place
No irrigation advice is shown	*Start irrigating below* has no standard value	After the first week, ask the grower and enter it (step 10)

More is in the runbooks (Project Manager Field Manual, PDF 02): R2 (a device does not answer) and R3 (a reading looks wrong).

Not yet proven

- **No soil probe has been read by this software, and no model is approved.** Every make is a first time: the test above is the proof, for that probe.
- Greecon says nothing about what a soil reading means for a crop. The limit is the grower's own, and Greecon never waters anything: it only reads.
- The prices are planning figures, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	

HW14

Temperature and humidity sensor

The temperature and humidity sensor reads the air at the field or in the greenhouse. The platform calls it the **Weather sensor**. It speaks Modbus RTU on the same data cable as everything else. **No model is approved yet,** so it qualifies by its maker's register table: its address is entered by hand, and its two readings are mapped by hand. Its most useful work is the frost alert, and that uses the grower's own number.

At a glance

WHAT IT DOES	Measures air temperature and humidity at the field. Unit 7, mapped by hand from the maker's register table.
HOW IT QUALIFIES	**Qualified by its register table.** An air temperature and humidity sensor that speaks Modbus RTU on RS-485, whose maker publishes its register table, and whose number and speed can be set.
SPECIFICATION	RS-485 Modbus, with a published register table
NUMBER ON THE CABLE	7. Set the maker's way, at the office, before the sensor is mounted.
FITTED BY	The project manager.
PLANNING ESTIMATE (EUR)	20 to 80. Not a quote.
USED IN	Kit F. The table under *At each kind of site* says what it does at each.

How it qualifies

It qualifies if:

- ☐ It speaks **Modbus RTU on RS-485,** on two data wires.
- ☐ **Its maker publishes its register table,** and you have it before you buy: the register of the temperature and of the humidity, the kind of number each is, and its scale.
- ☐ **The table says how a temperature below zero is given.** It is usually a signed 16-bit number.
- ☐ Its unit number and its speed can be set, and the maker says how: to a number of its own, and to 9600 baud, 8 data bits, no parity, 1 stop bit.
- ☐ It is made for where it will hang: outdoors, a sensor with a shield against sun and rain.
- ☐ Its supply voltage is one the kit has: read its label against the 24 V supply.

STOP **Do not buy a sensor whose register table you have not seen.** A sensor whose maker does not document its registers cannot be mapped honestly, so it cannot be used.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	NOT USED	None	None.
S Solar plant with grid connection	NOT USED	None	None.
F Farm or greenhouse with irrigation	IN THE KIT	1	One, at the field or in the greenhouse. It gives Air temperature and Humidity. The frost alert says nothing until the grower's own Frost alert at or below value is entered.
W Water tank and pump station	NOT USED	None	None.

What Greecon reads from it

Reading in Greecon	Unit	What is real	Standard alert
Air temperature	°C	-40 to 70	None. *Frost alert at or below* has **no standard value**
Humidity	%	0 to 100	None

The frost alert. Frost belongs to the crop: the same air is harmless to one and ruinous to another, and it depends on how far on the crop is. So nobody can set the limit for the grower, and until it is entered Greecon says nothing about frost. Once it is entered, an **Act now** alert opens when the air has been at or below it for two minutes: *Frost: the air at the field has fallen to the frost limit*. It can come in the night, so agree who is woken.

Settings, and where they are made

Setting	Value	Where
Unit number	**7.** Many sensors are delivered as 1, the same as another device. Set it the maker's way, with the USB-to-RS-485 adapter and no other device on the cable	At the office
Speed	9600 baud, 8 data bits, no parity, 1 stop bit, set the maker's way	At the office
Its label	The number, written on the sensor	At the office

Fitting

PROJECT MANAGER

- [ ] **In the shade,** a metre or two above the crop, away from walls and from the pump house. A sensor in the sun reads the sun, not the air.
- [ ] In a greenhouse: at the height of the crop, away from the heater, the vents and the door.
- [ ] It joins the data cable in the daisy chain: A to A, B to B, no branches (sheet HW03).
- [ ] Photograph where it hangs, and write the place down.

In the platform

The farm template makes the device **Weather sensor** with its two readings and no registers.

1. **Its address** (step 4, **Where each device is on the network**). The box cannot recognise the sensor, so its line says **Enter by hand**. Under the device open **Enter the address by hand** and fill in the **IP address** (the converter's: the water module's line shows it), the **Port** (502) and the **Unit number** (7). Press **Save**.
2. **Its registers** (step 5, **Which register holds which reading**). Under each reading press **Choose the register**. The register kind is **Input** for a manual's 3xxxx and **Holding** for 4xxxx, and the address counts from zero. Under how it becomes the reading choose **Multiply and add**: a sensor that reports tenths of a degree is multiplied by 0.1. Press **Save the register**.
3. **Map the temperature as a signed number (int16).** Mapped as an unsigned one, a frosty morning reads as a hot afternoon: -2 degrees becomes thousands.
4. **The frost limit** (step 10, **Alert limits agreed with the customer**). Ask the grower at what air temperature frost damages the crop, and in which weeks. Enter it as *Frost alert at or below*, between -30 and 15 degrees.

Write what you enter:

Reading	Register kind	Address	Kind of number	Multiply by
Air temperature			int16	
Humidity				

Acceptance test

At the office, on the desk:

- [ ] The sensor has its number (7), alone on the cable, and is labelled.
- [ ] Behind the converter, both registers are read from the box, and the numbers are the room's:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 7 --type input --address 1 --data-type int16
--scale 0.1
```

(The kind, the address, the kind of number and the scale are the maker's: these are an example.)

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
Weather sensor. Against a hand-held thermometer at the sensor. No tolerance is laid down: write both numbers and judge.	Plausible for the time of day, and close to the thermometer			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
A frosty morning reads as a hot afternoon, or as thousands of degrees	The temperature was mapped as an unsigned number	Map it as int16 (signed), with the scale of the sensor's table
The sensor does not answer	A wrong unit number (left as delivered), a number or an address entered wrongly, a speed that is not the cable's, A and B swapped, or no power	Check each, in that order. Test the register from the box against the maker's table
Temperature and humidity are swapped	The two registers are the other way round	Read the maker's table again
A reading is ten times too big	The scale	Under **Multiply and add**, multiply by the table's factor
The air reads far hotter than the thermometer at midday	The sensor is in the sun, or by a wall	Move it into the shade, away from walls
No frost alert came on a frosty night	The frost limit was never entered	Enter the grower's own number (step 10), and send a test message to the person who is woken

More is in the runbooks (Project Manager Field Manual, PDF 02): R2 (a device does not answer) and R3 (a reading looks wrong).

Not yet proven

- **No temperature and humidity sensor has been read by this software, and no model is approved.** Every make is a first time: the test above is the proof, for that sensor.
- The frost alert is tested in software. It has not yet been seen on a real frosty night.
- The prices are planning figures, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	

HW15

24 V power supply

The 24 V supply is the one clean place the kit's low-voltage parts take their power from: the converter, the two modules and the transmitters' loops, and the box if the box takes 9 to 36 V. It sits on the DIN rail in the enclosure, on the mains side, behind its own fuse or small breaker. Its mains side is the electrician's work. Its 24 V side is the project manager's.

At a glance

WHAT IT DOES	Powers the two modules, the loop transmitters and the converter from one place.
HOW IT QUALIFIES	**Qualified by specification.** Any 24 V DC supply for a DIN rail, of 30 to 60 W.
SPECIFICATION	30 to 60 W
FITTED BY	The electrician connects its mains side. The project manager connects the 24 V side to the converter and the modules.
PLANNING ESTIMATE (EUR)	25 to 50. Not a quote.
USED IN	Kits H, S, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

Any make qualifies that meets every line below.

It qualifies if:

- ☐ It gives **24 V DC,** fits a DIN rail, and is rated **30 to 60 W.**
- ☐ The mains voltage on its label is the site's.
- ☐ **Its power covers the kit.** Add up what the labels of everything it feeds say, and take a supply rated for that and half as much again. As a guide: 30 W is enough for the converter, the modules and the loops; take 60 W where the box or the 4G router shares the supply.
- ☐ Its data sheet says that its output is protected against a short circuit and an overload.
- ☐ It is a known make, with its safety approvals on the label, and the electrician who connects it accepts it. It is the part of the kit that is connected to the mains.

STOP

Do not use a plug-in charger or an open supply with bare terminals in the enclosure. **The project manager never connects the mains side:** that is the electrician's work, and it has its own fuse or small breaker.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1	One, in the enclosure. It feeds the two modules, the transmitters' loops and the converter, and the box if the box takes 9 to 36 V.
S Solar plant with grid connection	IN THE KIT	1	One. With no module and no loop, it feeds the converter, and the box if the box takes 9 to 36 V.
F Farm or greenhouse with irrigation	IN THE KIT	1	One, as at a hotel: the modules, the loops and the converter.
W Water tank and pump station	IN THE KIT	1	One, as at a hotel: the modules, the loops and the converter.

What it feeds

It feeds	The part takes	When
The converter	9 to 36 V DC	Always
The 4 to 20 mA module	7 to 36 V DC	Kits H, F and W
The pump contact module	The voltage on its label	Where there is a pump to watch
Each transmitter's loop	The loop's current, up to 20 mA	A two-wire transmitter takes its power from the loop
The box	9 to 36 V DC	Only a box that takes 9 to 36 V. A box that takes 12 V keeps its own supply, or a small adapter from 24 V to 12 V
The 4G router, a soil probe, the weather sensor	The voltage on its label	Only if its label allows 24 V

Read every label before a part is connected to 24 V. A part made for 5 V or 12 V is destroyed by 24 V.

Settings, and where they are made

Setting	Value	Where
Output voltage	24 V. Many supplies have a small screw that trims it: leave it at 24 V	Measured with the multimeter, at the office

Fitting

ELECTRICIAN connects its mains side. **PROJECT MANAGER** connects the 24 V side.

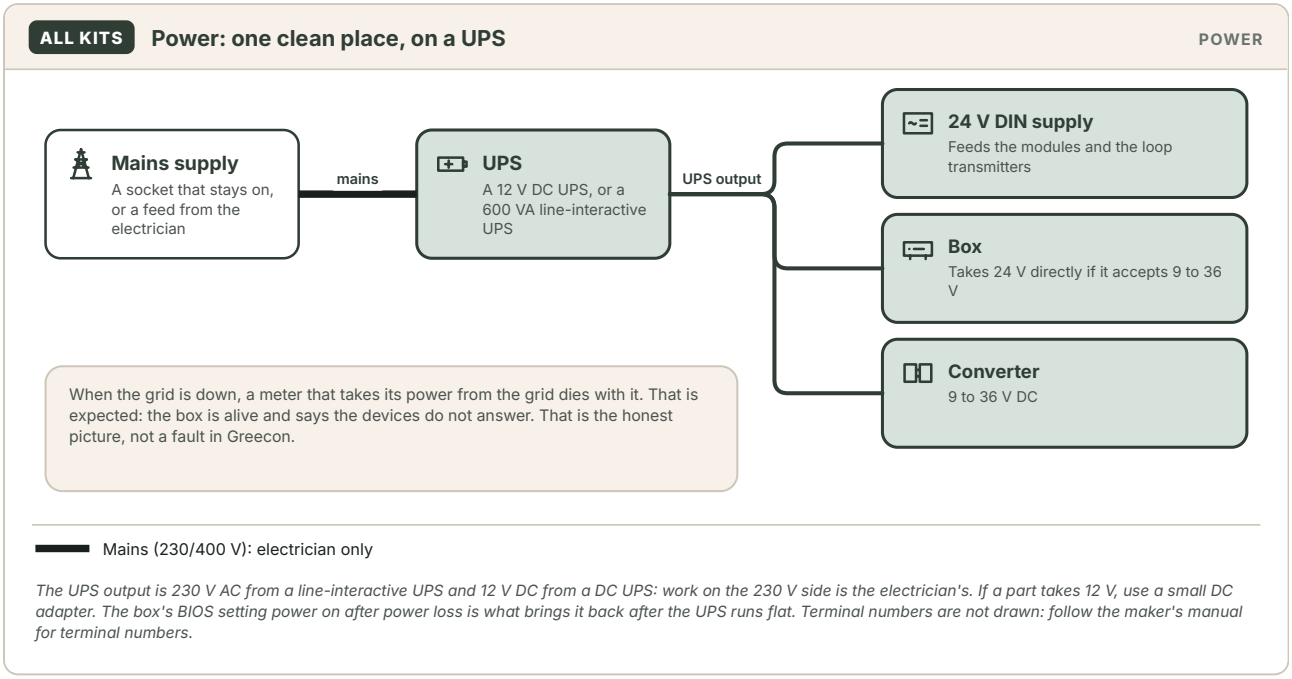


Figure 4-4. Power: everything from one clean place, on a UPS.

- ☐ It sits on the DIN rail, on the mains side of the enclosure, with the air around it that its manual asks for.
- ☐ **The mains comes in through its own fuse or small breaker.** The electrician connects it.
- ☐ Where the site has a mains UPS, the supply is behind it, so that the modules and the loops keep working in a short power cut (sheet HW16).
- ☐ On the 24 V side, plus and minus go to the terminal blocks, and from there to each part. Plus and minus are checked against each part's label before the supply is switched on.
- ☐ Each transmitter's loop has the supply in series with the module's input, as the module's manual draws it (sheet HW08).
- ☐ Number every wire at both ends.

In the platform

The supply is not in the platform, and Greecon does not read it. What shows that it is working is everything it feeds: the modules answer, and the transmitters read 4 mA or more.

Acceptance test

At the office, on the desk. The electrician makes up the office's bench lead: the project manager never wires mains.

- [ ] With nothing connected, the multimeter reads 24 V at the output, within a volt.
- [ ] With the whole kit connected and running for an hour, it still reads 24 V, and the supply is not hot.
- [ ] The box finds every device (`greecon-edge find --host 192.168.1.50`).

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
The 24 V side. With everything connected, measure at the supply's output and at the part farthest from it.	24 V at both, within a volt			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
Everything on the 24 V side is dark	No mains at the supply: its fuse or breaker, or the UPS in front of it	The electrician checks the mains side. You do not open it
The supply's light is on, and one part is dark	A loose 24 V wire, or plus and minus the wrong way round at that part	Switch the supply off, and check the two wires against the part's label
The 24 V sags, or the supply clicks on and off	Too much on the supply, or a short circuit on the 24 V side	Disconnect the parts one at a time to find it. A 30 W supply that also feeds the box or the router is changed for 60 W
Every transmitter reads below zero at once	The loops have lost their 24 V	As the first line. One transmitter alone is a wire: see its own sheet

More is in the runbooks (Project Manager Field Manual, PDF 02): R2 (a device does not answer).

Not yet proven

- What each part really draws has not been measured: the guide above is from labels. Measure it at the rehearsal of the first kit, and write the figure down.
- The prices are planning figures, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	

HW16

UPS

The UPS keeps the box and the converter running through a short power cut. With it, a power cut looks like what it is: the box is alive and says that the meters do not answer. Without it, the box goes silent with everything else, and nobody can tell a power cut from a broken box until the power returns. It is not there to run the site, and it does not make Greecon a safety system.

At a glance

WHAT IT DOES	Keeps the box and the converter up through short power cuts and long enough to say that the grid is down.
HOW IT QUALIFIES	**Qualified by specification.** A 12 V DC UPS for the box and the converter, or a 600 VA line-interactive UPS.
SPECIFICATION	A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS
FITTED BY	The project manager places it. The electrician connects anything that is wired to the mains.
PLANNING ESTIMATE (EUR)	60 to 150. Not a quote.
USED IN	Kits H, S and W. The table under *At each kind of site* says what it does at each.

How it qualifies

Either of two kinds qualifies. Which one is bought decides what keeps working in a power cut.

Kind	It feeds	In a power cut
12 V DC UPS	The box and the converter, at 12 V. The box must take 12 V; the converter takes 9 to 36 V	The box and the converter stay up. The 24 V supply is on the mains, so the modules and the transmitters go dark, and the box says that they do not answer
600 VA line-interactive UPS (mains in, mains out)	The box's own supply and the 24 V supply, which feeds the converter, the modules and the loops	The whole kit stays up. The tank and the pressure are still read. Meters powered from the grid go dark

It qualifies if:

- ☐ It is one of the two kinds above, and what it must feed fits it: read the labels.
- ☐ **It comes back by itself.** After it has run flat and the mains returns, its output comes on again with nobody pressing a button. Test this on the desk.

- [ ] Its battery can be changed, and the date of the battery is written on it. A UPS battery wears out in a few years.
- [ ] A mains UPS has enough sockets that are backed by the battery. Many have some sockets that only protect against surges: the kit goes into the backed ones.

STOP

A UPS is for short cuts. Nothing but the kit is plugged into it. When the grid is down, a meter powered from the grid is dark, and Greecon says that it does not answer. That is the honest picture ("we cannot tell"), not a fault.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1	One. It keeps the box and the converter running through a short power cut. A 600 VA UPS keeps the 24 V supply up too, so the tank and the pressure are still read.
S Solar plant with grid connection	IN THE KIT	1	One. It keeps the box and the converter running. The meters are powered from what they measure: in a grid cut they go dark, and the box says that they do not answer.
F Farm or greenhouse with irrigation	NOT USED	None	Not in the standard list for this kit. Decide at the survey whether this site gets one, and write the decision down.
W Water tank and pump station	IN THE KIT	1	One. A 600 VA UPS is the better choice here: it keeps the 24 V supply up too, so the tank and the pressure are still read while the power is down.

At a site with water (a hotel, a water station), the 600 VA kind is the better choice: the tank is still read while the power is down, which is when it matters most.

What it is for

- **To tell a power cut from a broken box.** The box stays alive and reports. On the site's page the devices powered from the grid do not answer, and the box does.
- **To keep the box from being cut off in the middle of its work.** A box on a real disk survives that, but it is kinder not to test it every week.
- **Not to keep the internet up.** The box can only report while the site's router has power. Where the router is Greecon's 4G router, it goes behind the UPS. Where it is the customer's, ask whether it has a UPS of its own, and write the answer down. If it has none, the box keeps the readings and sends them when the internet returns.
- **Not for long cuts.** When the UPS runs flat the box stops. The BIOS setting "power on after power loss" (sheet HW01) is what brings it back when the power returns.

Fitting

PROJECT MANAGER places it. ELECTRICIAN connects anything that is wired to the mains and not plugged in.

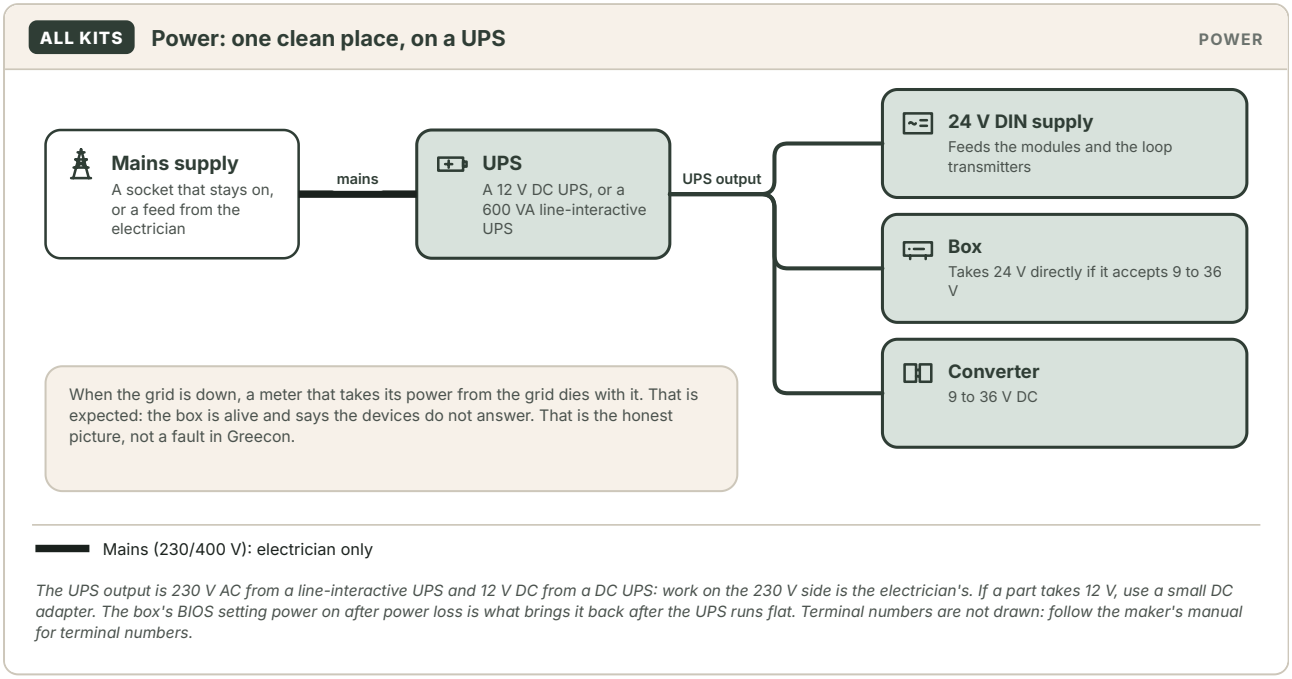


Figure 4-4. Power: everything from one clean place, on a UPS.

The drawing shows a mains UPS, which feeds all three. A 12 V DC UPS feeds the box and the converter only: the 24 V supply is then on the mains.

- ☐ A mains UPS stands on a shelf beside or below the enclosure, with the air around it that its manual asks for, plugged into a socket that stays on.
- ☐ A 12 V DC UPS sits on the rail or beside it. The box and the converter take their power from its 12 V output.
- ☐ The kit is plugged into the sockets that the battery backs.
- ☐ The 4G router, where the kit has one, is behind the UPS too.
- ☐ **Label it** with the date of its battery. Nothing else is plugged into it.

In the platform

The UPS is not in the platform, and Greecon does not read it. What shows it is working is the box: during a power cut it stays **Online** for as long as the site's internet has power.

Acceptance test

At the office, on the desk, with the kit running:

- ☐ Pull the UPS's mains plug. The box and the converter stay on. With a mains UPS, the modules stay on too.

☐ Leave it until the UPS gives up, and **write down how many minutes that took.**

☐ Give it mains again. Its output returns by itself, the box starts by itself, and the box reports within three minutes.

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
A power cut on the kit. Pull the UPS's plug for two minutes. Only the kit's power, never the site's.	The box stays Online. With a mains UPS, the readings of the water module keep arriving			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
The box went Offline in a short power cut	The UPS's battery is worn out; the box is in a socket that only protects against surges; or the site's router lost its power	Pull the UPS's plug on a quiet day and watch. Check which socket the box is in. Ask about the router
After a long cut the box stayed off	The box's BIOS is not set to power on after power loss, or the UPS does not come back by itself	Set the BIOS (sheet HW01). Run the UPS flat on the desk and see that it returns
The UPS beeps, or shows a battery fault	Its battery is worn out	Change the battery or the UPS, and write the new date on it
The meters do not answer during a cut	They are powered from the grid	Nothing. It is expected, and it ends by itself

More is in the runbooks (Project Manager Field Manual, PDF 02): R1 (a box is offline).

Not yet proven

- **How long a UPS holds a kit has not been measured.** That is why the minutes are timed at the office and written down.
- No kit has yet been through a real power cut at a site.
- The standard list for a farm kit has no UPS, although every kit is meant to be powered on one. The decision is taken at the survey, and written down.
- The prices are planning figures, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	

HW17

Enclosure and fittings

The enclosure is the one box on the wall that holds the kit: the fuses and the 24 V supply on its mains side; the converter, the modules and the terminal blocks on its low-voltage side; and the box itself, if there is room. It keeps dust and water out, it keeps mains and low voltage apart, and, labelled properly, it lets the next person understand the kit without asking anyone.

At a glance

WHAT IT DOES	Holds the converter, the modules, the supply, the fuses, the terminal blocks and, if there is room, the box.
HOW IT QUALIFIES	**Qualified by specification.** Any enclosure of IP54 (IP65 outdoors) with a DIN rail, terminals, fuses or small breakers and glands, large enough for the kit.
SPECIFICATION	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands
FITTED BY	The project manager mounts it and the low-voltage parts in it. The electrician connects the mains side.
PLANNING ESTIMATE (EUR)	80 to 150. Not a quote.
USED IN	Kits H, S, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

Any make qualifies that meets every line below.

It qualifies if:

- ☐ It is rated **IP54** indoors, and **IP65** outdoors or where it can get wet.
- ☐ **It is large enough.** Lay the kit's parts side by side on a desk and measure: the DIN rail takes them all, with room for the terminal blocks and for air around the supply.
- ☐ It has a DIN rail, terminal blocks for the data cable and for 24 V, and fuses or small breakers: one for the supply's mains side and, where the electrician places them here, those of the meters' voltage taps.
- ☐ It has a cable gland for every cable that enters, at the bottom.
- ☐ Its door closes with a key or a tool.
- ☐ At a site with a 4G router: it is plastic, or the router's aerial can be led outside. A steel enclosure weakens the signal.

STOP

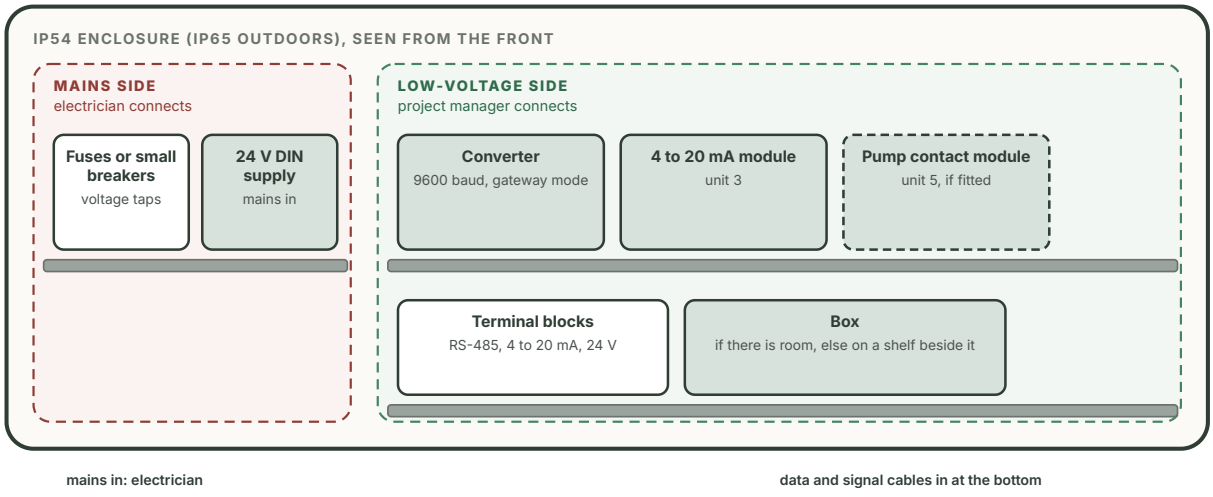
Mains and low voltage are kept apart in it, and the mains side is the electrician's. The project manager mounts the enclosure and connects the low-voltage side, and never connects, opens or alters the mains side.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1	One: the fuses and the 24 V supply on the mains side; the converter, the 4 to 20 mA module and the pump contact module on the low-voltage side; the terminal blocks and, if there is room, the box.
S Solar plant with grid connection	IN THE KIT	1	One, the smallest: the fuses and the supply, the converter, the terminal blocks and, if there is room, the box. The meters are in the distribution board, not here.
F Farm or greenhouse with irrigation	IN THE KIT	1	One, in the pump house: as at a hotel, with the 4G router on the rail. The probes and the weather sensor are in the field, on the data cable.
W Water tank and pump station	IN THE KIT	1	One: as at a hotel. The optional pump meter sits in the board that feeds the pump, not in this enclosure.

The order of the parts on the rail, for each kind of site. It is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure.

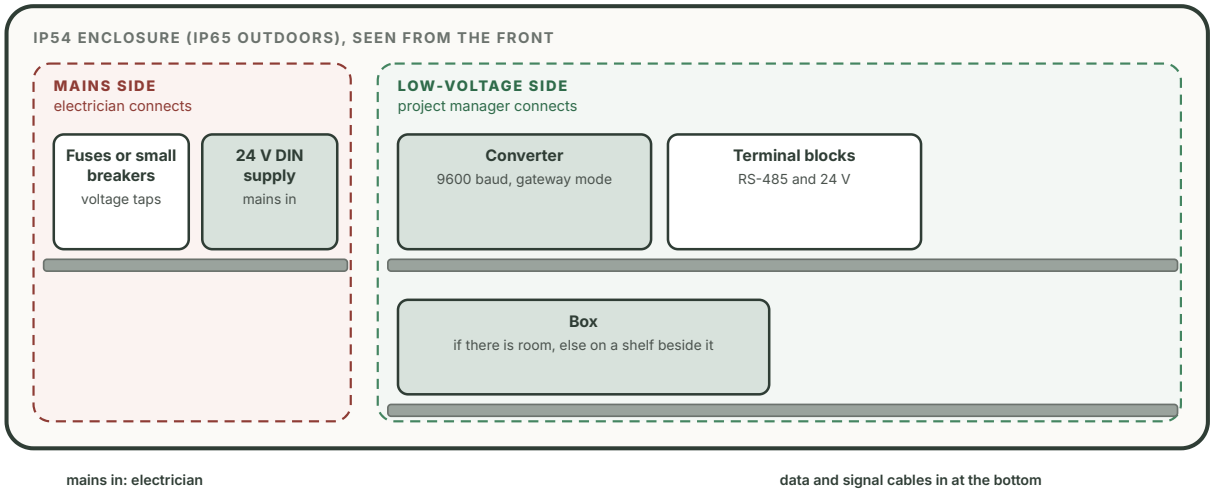
KIT H **The enclosure: recommended order on the DIN rail** ENCLOSURE



The meters are not in this enclosure: they sit in the distribution board at the point they measure.
The order is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure H-4. Kit H. The enclosure: the recommended order of the parts on the DIN rail, mains side on the left.

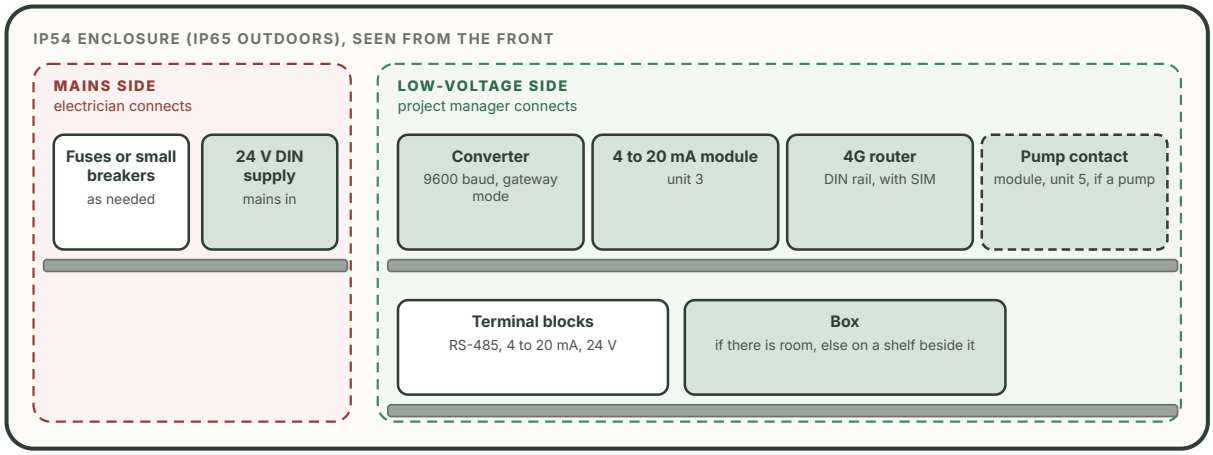
KIT S **The enclosure: recommended order on the DIN rail** ENCLOSURE



The meters are not in this enclosure: they sit in the distribution board at the point they measure.
The order is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure S-4. Kit S. The enclosure: the recommended order of the parts on the DIN rail, mains side on the left.

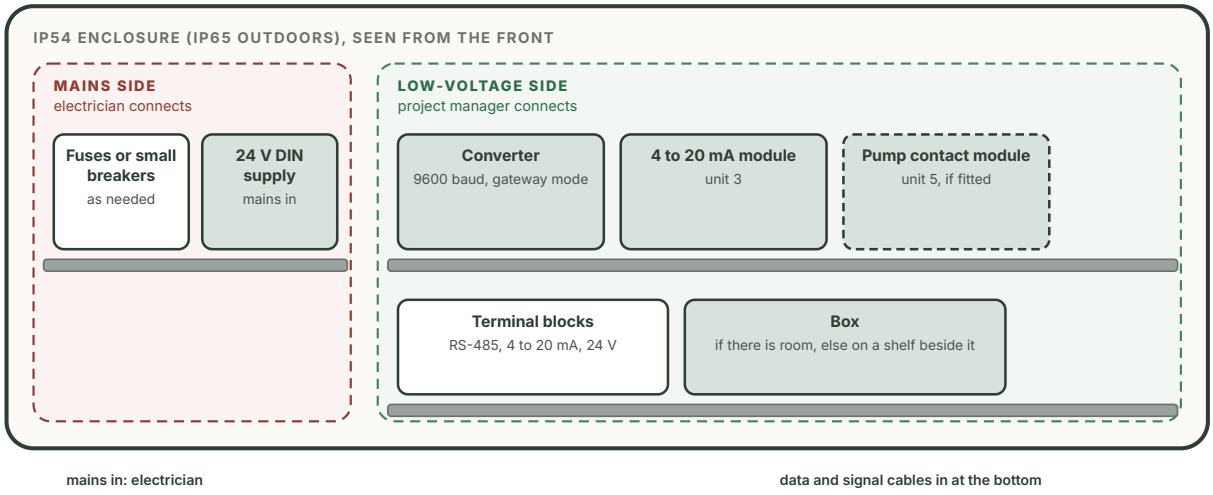
KIT F **The enclosure: recommended order on the DIN rail** ENCLOSURE



The probes and the weather sensor are in the field, on the data cable. Check the signal where the router sits.
The order is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure F-4. Kit F. The enclosure: the recommended order of the parts on the DIN rail, mains side on the left.

KIT W **The enclosure: recommended order on the DIN rail** ENCLOSURE



The optional pump meter sits in the board that feeds the pump, not in this enclosure.
The order is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure W-4. Kit W. The enclosure: the recommended order of the parts on the DIN rail, mains side on the left.

What is labelled

What	Its label
Each device	Its unit number, as set at the office
The converter	Its settings: gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit, port 502, and its address
Every wire	A number, at both ends
The UPS	The date of its battery
Inside the lid	The site name and the date

Never write a code or a password on the box, the enclosure or a label.

Fitting

PROJECT MANAGER mounts the enclosure and the low-voltage parts in it. ELECTRICIAN connects the mains side.

- ☐ **The place:** a dry wall, away from heat, where the door can be opened fully and the labels read. Outdoors, out of the sun where that is possible.
- ☐ **Cables come in at the bottom,** each through its own gland. Unused holes are plugged.
- ☐ **Mains on one side, low voltage on the other,** as the drawing shows, and the data cable away from power cables.

- [ ] The box goes inside if there is room, otherwise on a shelf beside it.
- [ ] **Photograph the inside before closing it,** with every label readable.

In the platform

The enclosure is not in the platform. What the platform keeps is what is written on its labels: each device's unit number and the converter's address, on the site's install card.

Acceptance test

At the office, before the kit leaves:

- [ ] The parts are mounted on the rail in the order of the drawing, and the door closes.
- [ ] Every device carries its unit number, and the converter its settings.
- [ ] The rehearsal on the desk was done with the parts in the enclosure, as they will be on site.

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
Closed up. Look before the door is shut.	Glands tight, unused holes plugged, mains and low voltage apart, every wire numbered at both ends, the labels in place, and the photograph taken			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
Water or dust inside	A gland that was not tightened, an unused hole, or IP54 where IP65 was needed	Tighten and plug. Outdoors, or where it is hosed down, change it for IP65
The box restarts on hot days	The box and the supply in a small closed enclosure, in the sun or beside a heat source	Move the box to a shelf beside the enclosure, or move the enclosure into the shade
Nobody can tell which wire is which	The numbers were left off	Number both ends now, against the photograph, and take a new photograph
The 4G router hears nothing with the door shut	A steel enclosure	Lead the aerial outside, as the router's manual allows (sheet HW04)

Not yet proven

- **No kit has yet been assembled in an enclosure.** The order on the rail is a recommendation. The first one is assembled at the office and photographed, and this sheet is corrected from it.
- The prices are planning figures, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	